Priced Out: Housing Costs, Generational Wealth Gaps, and Youth Welfare

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Abstract

Later birth cohorts of Americans leave home later and form families later, and they also own less and live in smaller homes. We ask how much of these housing outcomes the economic environment alone can explain. We build a quantitative life-cycle model in which renters and owners choose among three quality tiers and all cohorts share the preferences we estimate on the 1940 cohort, which rules out changes in tastes as an explanation. Even so, the model reproduces the fall in ownership at ages 25 to 34 from the 1940 to the 1980 cohort, close to the observed 22 percentage points. It also reproduces the shift of later cohorts toward smaller homes at every age, so ownership alone understates what later cohorts have lost. In a counterfactual, we give the 1980 cohort the house prices, interest rates and earnings of the 1940 cohort and split the difference across the three. House prices account for the largest part of the ownership gap at every age. The welfare loss answers to a different force. The median household in the 1980 environment would need 58.6 percent more lifetime non-housing consumption to be as well off as in the 1940 environment, and earnings contribute somewhat more to this than prices. Which lever matters therefore depends on the goal. The loss falls hardest on households that begin with low earnings and little wealth. Housing also takes a growing share of the budget, leaving each later cohort less for everything else in the years when the big commitments are made.

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I. Introduction

The US homeownership rate has stayed close to 65 percent for six decades. Behind that stable aggregate, each birth cohort has owned less than the one before it. In the Panel Study of Income Dynamics (PSID), more than 70 percent of households in the 1940 cohort owned a home at age 35, against about 60 percent in the 1960 cohort and about 50 percent in the 1980 cohort. Later cohorts rent for longer and own less at every age we observe.

Owning less and renting for longer carry costs through the life cycle. These costs fall on all households, but most heavily on those with lower incomes and on the public programs that support them. Young households without savings for a down payment find it hardest to buy (Acolin et al., 2016). Home equity is the largest part of most households' wealth outside pensions (Goodman and Mayer, 2018), and those who do not buy do not fully make up for it with financial assets (Paz-Pardo, 2024). The consequences do not end in early adulthood. Housing costs weigh increasingly on older renters, and most of those who qualify for rental assistance do not receive it (Joint Center for Housing Studies of Harvard University, 2023, 2025). As later cohorts, who rent for longer, reach old age, this pressure will grow. Reading the rise in renting among the young as a choice therefore understates its cost, and sets aside the possibility that housing costs shape family formation (Dettling and Kearney, 2014).

How much of the fall in ownership can the economic environment explain on its own? Suppose the 1980 cohort had faced the house prices, interest rates and earnings of the 1940 cohort. How much more would it own, how much larger would its homes be, and how much bette ⌞r off would it be? And which households would gain the most? The comparison spans a large change in conditions: the 1940 cohort entered the housing market in the mid-1960s, while the 1980 cohort entered in the mid-2000s, after house prices had risen well ahead of incomes. We answer these questions with a quantitative life-cycle model in which all cohorts share the same preferences, which rules out changes in tastes as an explanation and lets us ask whether the environment alone can account for the facts.

Model and measurement Households enter the model at age 25 and each year choose consumption, saving, whether to rent or own, and the quality of their home, facing income and mortality risk and borrowing constraints. Homes come in three quality tiers, defined by the number of rooms in the PSID, and renters choose among the same tiers as owners. This lets the model speak not only to whether households own but also to what they live in. Each cohort faces the house prices and interest rates of its own time and an earnings process estimated on its own family earnings in the PSID, which captures lower median

earnings and wider dispersion at entry, and fewer two-earner couples, in later cohorts. We attribute the cross-cohort gap to these three channels with a Shapley decomposition (Shorrocks, 2013), also used by Paz-Pardo (2024), which averages each channel's effect over every order in which the channels can be switched.

Ownership alone shows only part of the loss. A household can still own and be worse off, because it buys a smaller home or pays more for the same one. We therefore measure welfare by the ex ante compensating variation at age 25: the percentage increase in lifetime non-housing consumption that would give a household in the 1980 environment the expected utility it would have had in the 1940 environment. We split this loss across the same three channels and by income and wealth at age 25.

Results First, house prices account for the largest part of the fall in ownership. At ages 25 to 34 the gap between the 1940 and 1980 cohorts is 22.0 percentage points in the data and 25.8 in the model, at the edge of the sampling error of the data gap. In the 1940 environment the 1980 cohort would own more at every age, and the difference widens with age. House prices make the largest contribution at every age, and their contribution grows with age as cheaper housing accumulates into wealth through earlier entry, lower mortgage burdens and moves to larger homes. Earnings contribute little at ages 25 to 34 and more with age, as a shortfall in income becomes a shortfall in wealth, but even in old age their contribution stays well below that of house prices.

Second, later cohorts not only own less but also live in smaller homes. At ages 25 to 34 the share of households in the lowest quality tier rises across cohorts from 31 to 61 percent in the data and from 37 to 65 percent in the model. The model also reproduces the 1940 cohort's move into larger homes in middle age and the ordering in which later cohorts stay in lower tiers at every age. Since the model has no marriage or fertility decision and no variation in household size, the pattern it produces comes from house prices relative to early-career earnings alone.

Third, the welfare loss is large. The median household in the 1980 environment would need 58.6 percent more lifetime non-housing consumption to be as well off as in the 1940 environment, a figure that combines later ownership, smaller homes and lower consumption early in life. Later cohorts spend more of their budget on housing at entry, among both renters and owners, and have less left for everything else. In relative terms, the data point the same way. In the CEX, median non-housing consumption of young households has barely grown since the early 1990s, while that of households aged 65 and over has risen steadily. Earnings and house prices carry most of the welfare loss, earnings somewhat more. House prices set the bar for buying a home, but once tenure is chosen,

earnings matter most for how much is left to live on, year after year. Which lever matters therefore depends on the goal.

Fourth, the cost is unevenly shared, and its timing differs across the income distribution. At prime age (45 to 54), the 1940 environment would raise ownership by 35 percentage points for the bottom fifth of lifetime income, but by 7 for the top fifth and by 4, 3 and 0.2 for the top 10, 5 and 1 percent. High earners do lose, but mainly through delay: in the 1940 environment the top 10 percent would have owned 22 points more at ages 25 to 29, a gap that narrows to 4 points by their mid-40s. For the bottom fifth the gap instead widens with age, from 15 points at ages 25 to 29 to 35 points at 45 to 54. The welfare loss follows the same order: households in the bottom fifth of age-25 income would need 74 percent more lifetime consumption against 39 percent for the top 5 percent, and sorted by age-25 wealth the range runs from 94 to 35 percent.

Related literature Our model builds on partial equilibrium life-cycle models of tenure and housing demand with income risk, transaction costs and down-payment constraints (Díaz and Luengo-Prado, 2008; Attanasio et al., 2012; Bajari et al., 2013), and on the idea of a property ladder (Ortalo-Magné and Rady, 2006). In our model renters and owners choose among the same three quality tiers, which lets us ask whether the environment alone reproduces the life-cycle path of housing quality and the shift of later cohorts toward smaller homes, and lets us value what these changes mean for welfare. On welfare, Li and Yao (2007) show that a permanent rise in house prices improves the welfare of middle-aged and older owners but leaves young owners and renters worse off, and Fagereng et al. (2025) find in Norwegian data that rising asset prices redistributed from the young, who are net buyers of housing, to the old. We compare instead the whole paths of house prices, interest rates and earnings that different cohorts faced.

Other studies link ownership among the young mostly to household formation, earnings and credit. Changes in the likelihood of marriage and divorce moved ownership among young singles and couples in opposite directions, and singles who expect to marry may delay buying (Chang, 2024). Fisher and Gervais (2011) attribute three fifths to four fifths of the fall in ownership at ages 25 to 44 between 1980 and 2000 to later marriage and higher earnings risk, Haurin and Rosenthal (2007) estimate that lower headship reduced ownership in the early and mid twenties by 3 to 5 percentage points between 1970 and 2000, and Paz-Pardo (2024) attributes a large part of the decline across generations to changes in earnings dynamics. Student debt also delays ownership (Mezza et al., 2020), and the terms of mortgage credit can move it a great deal, as they did in the boom of 1994 to 2005 (Chambers et al., 2009). These explanations complement ours. The cohorts we

compare entered the market before and after a large rise in house prices relative to incomes, and a comparison of steady states with fixed house prices, as in Fisher and Gervais (2011), cannot capture that rise.

Composition matters, but it is not the whole story. Between the 1940 and 1980 cohorts, ownership at ages 25 to 34 fell from 65 to 49 percent among couples and from 25 to 17 percent among singles, and couples at age 30 had almost the same median income in both cohorts. It also cuts the other way. The share of 25-year-olds who head a household or are married to a head fell from 87 to 70 percent, and young adults who still live with their parents drop out of a household-based measure. Both the data and the model count households, so both likely understate the fall per person. Household formation and household size are not part of our baseline model. When we adjust consumption for each cohort's household size with the OECD equivalence scale as a robustness check, the model's ownership gap narrows by 2 to 5 percentage points across ages, but our main results continue to hold.

Section II describes the data and three facts, the widening wealth gap between age groups, the weight of housing in household net worth and spending, and later entry into ownership across cohorts. Section III presents the model, and Section IV estimates its preference parameters on the 1940 cohort by the method of simulated moments. Section V reports the decomposition of ownership and housing quality, non-housing consumption and housing cost burdens, and the welfare loss. Section VI examines robustness to income projections, house price paths and household size, and Section VII concludes.

II. Data

II.A. Sources

Our analysis draws on three U.S. household surveys. The Panel Study of Income Dynamics (PSID) has followed the same families since 1968, making it well suited for lifecycle patterns in tenure, income, and housing. The Survey of Consumer Finances (SCF), conducted every three years by the Federal Reserve, provides detailed cross-sectional snapshots of household balance sheets. The Consumer Expenditure Survey (CEX), collected by the Census Bureau for the Bureau of Labor Statistics, tracks household spending on housing and non-housing goods. Together the three datasets cover lifecycle dynamics, wealth heterogeneity, and consumption patterns across cohorts. Table 1 summarises the surveys.

Table 1. Survey Data Sources

	PSID	SCF	CEX
Sponsor	U. of Michigan / NSF	Federal Reserve	BLS / Census Bureau
Available	1968–2023	1989–2022	1984–2024
Frequency	Annual (biennial post-1997)	Triennial	Annual
Structure	Panel	Repeated cross-section	Rotating panel
Sample size	∼10,000 families	∼6,500 families	∼7,000 households
Oversampling	Low-income	High-wealth	None
Use in paper	Tenure, income, wealth	Wealth distribution	Consumption

Notes: PSID sample refers to SRC and SEO components combined. SCF oversamples wealthy households using IRS Statistics of Income data. CEX microdata (PUMD) available 1980–2024; annual published tables from 1984.

II.B. Sample and Cohort Definitions

We define three birth cohorts: the 1940 cohort (born 1940–49), the 1960 cohort (born 1960–69), and the 1980 cohort (born 1980–89). All monetary values are in 2022 USD (deflated using the CPI). PSID tracks all three cohorts continuously from age 25, though the 1940 cohort's early waves (pre-1968 birth years) are reconstructed from retrospective responses.[1]

Descriptive statistics at ages 25–34, where all three cohorts overlap, are reported in Appendix A. Three changes stand out. First, the homeownership rate fell from 53% to 31%. Second, real median family income declined from $72,000 to $55,000, reflecting losses concentrated in the lower half of the distribution rather than deteriorating human capital. The median is the relevant statistic here: the mean falls far less, so the loss is concentrated in the lower half of a distribution that has widened.

Third, the share currently married fell from 52% to 40%, female headship nearly doubled, and the share with any children fell from 69% to 41%, changes that interact with both housing demand and mortgage eligibility.

II.C. Motivating Facts

Fact 1: The wealth gap between age groups has widened Figure 1 plots mean household net worth by age group (the Survey of Consumer Finances). Panel A shows that net worth rises steeply with age, peaks in the 55–64 and 65–74 groups, and has grown sharply for older households. Panel B shows the ratio of mean wealth for older groups relative to younger ones: the 65–74 / 25–34 mean ratio rose from roughly 5 in 1989, with a pronounced spike around 2010–2013 as the post-crisis recovery disproportionately benefited asset-rich

[1]Where early-age observations for the 1940 cohort are insufficient, we use the extended 1940 cohort (born 1940–55) or the 1950 cohort (born 1950–59); substitutions are noted where relevant.

older households.[2]



Figure 1. Net Worth by Age Group and Wealth Gap Ratios, 1989–2022

Notes: SCF 1989–2022. All values in 2022 USD. Survey-weighted means.

The macroeconomic implication is that today's young cohorts compete for assets against an older generation whose wealth has no historical precedent. When that wealth is concentrated in supply-constrained assets such as housing, it pushes up the prices that determine young households' access to ownership. The model below takes these price dynamics as given and examines how cohorts entering the housing market at different times along this trajectory differ in homeownership outcomes and housing quality. We trace the consequences for their lifetime welfare and for inequality across and within cohorts.

Fact 2: Housing dominates net worth and expenditure across the distribution Figure 2 Panel A shows that housing constitutes the majority of net worth for middle-income households (D5–D7), where the mean ratio exceeds one, implying that mortgage debt pushes net worth below house value. For low- and high-income deciles the ratio is below one but remains the largest single asset. Panel B shows that housing expenditure absorbs

[2]The ratio narrows at the very end of the sample, in 2022, when net worth for the youngest group rose unusually sharply. Part of this reflects a shift in who is sampled among the young, together with the Federal Reserve's 2022 redesign of the SCF's racial and ethnic oversample, so we read the single-wave narrowing as provisional until the next survey confirms it.

over 40% of income for renters in the bottom two deciles, well above the 30% cost-burden threshold, while owners spend roughly 10% regardless of income decile.



Figure 2. Houses to Net Worth and Housing expenditure to Income by Income decile

Notes: Panel A: mean (dark blue) and median (light blue) house-value-to-net-worth ratio, SCF 2022; renters included with a ratio of zero, negative-net-worth households excluded. Panel B: weighted median housing-expenditure-to-pre-tax-income ratio, CEX 2024. Deciles of pre-tax income (> 0), pooling renters and owners.

Fact 3: Younger cohorts enter homeownership later Figure 3 traces homeownership rates by single year of age for the three PSID cohorts. The 1940 cohort reached roughly 80% ownership by age 55. The 1960 cohort follows a similar path at a slightly lower level. The 1980 cohort, observed only through its early 40s, lags behind where the older cohorts stood at the same ages: at age 35 its ownership rate is roughly 10 percentage points below the 1960 cohort's and 20 percentage points below the 1940 cohort's, consistent with the deterioration in affordability documented above.

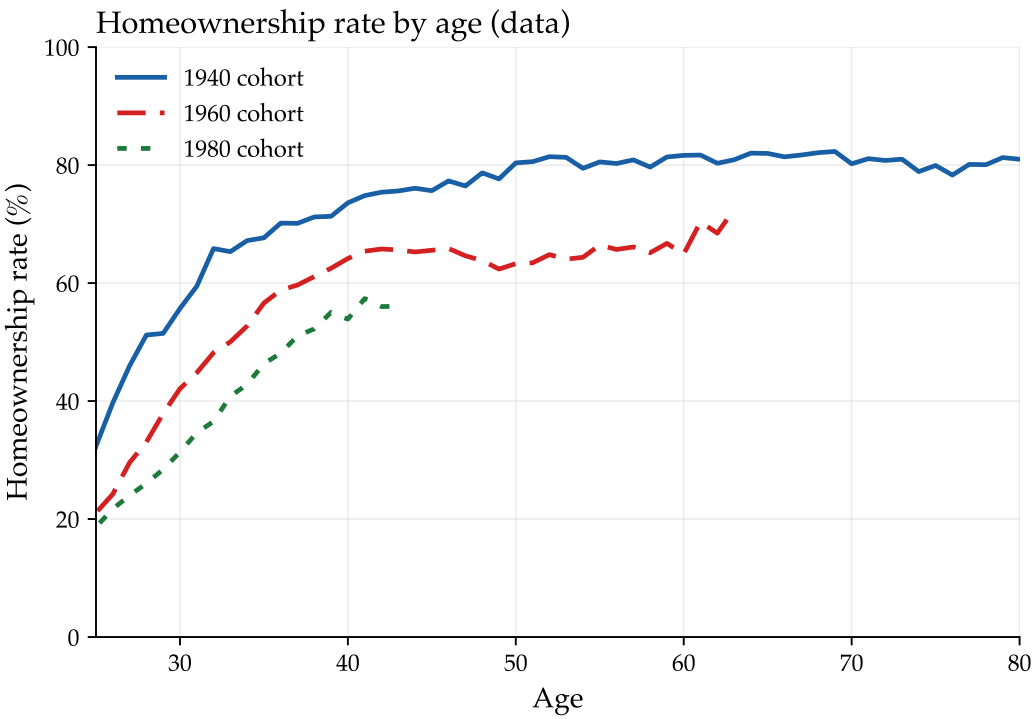


Figure 3. Homeownership Rate by Age and Birth Cohort

Notes: PSID, heads of household. Cohorts: 1940 (born 1940–49), 1960 (born 1960–69), 1980 (born 1980–89). Weighted using PSID longitudinal family weights.

Homeownership matters here on two counts: it is the main vehicle through which most households build wealth, and it insures against future rent growth. On both margins, the 1980 cohort is participating less at the ages when earlier cohorts entered. Whether this reflects a delay or a permanent shift, whether the cohort eventually catches up or carries lower ownership through the rest of the life cycle, is an open question, and the distinction matters. If more households reach old age as renters, they must cover market rents out of labour and pension income long after earnings have peaked, and the welfare consequences could be substantial. Supply-side reform may ease the underlying price pressure, and zoning liberalization has gained policy momentum in recent years;[3] but new supply responds slowly, varies widely across cities. As the recent swings in multifamily construction suggest, it appears sensitive to costs, rates, and regulation.[4] This paper does

[3]Oregon's House Bill 2001 (2019) was the first statewide measure of its kind, requiring cities above 10,000 residents to allow duplexes on lots zoned for single-family housing; Minneapolis eliminated single-family-only zoning citywide under its 2040 Comprehensive Plan (effective 2020); California's SB 9 (2021) mandated ministerial approval of duplexes and lot splits on single-family parcels; and Montana's 2023 reform package legalized duplexes and accessory dwelling units statewide. Several of these measures have faced legal challenges.

[4]Multifamily housing starts reached roughly 530,000 units in 2022, their highest level since the mid-1980s,

not model the supply side. It treats the observed path of house prices as given, and asks what entering the housing market at different points along that path has meant for successive cohorts: for tenure, for the housing they attain, for welfare, and for inequality across and within cohorts.

Median net worth among households aged 55 and above has risen substantially over the past decades, while that of households under 35 has stagnated. Younger cohorts now inhabit an economy where older age groups hold far greater wealth than their predecessors did at equivalent ages. In a world of inelastic housing supply, this wealth concentration pushes prices upward. Housing stands apart from other assets. It is essential consumption and the single largest expenditure item in household budgets. For homeowners, it is also the primary vehicle for wealth accumulation. The result is a widening squeeze for younger cohorts. Entry into homeownership comes later and reaches lower peak rates. Many cohort members remain renters longer into their lives and bear persistently high housing-to-income ratios while building no housing equity. At the very life stage when major commitments are made (independent household formation, childbearing, human-capital investment, precautionary saving), housing absorbs a growing share of available resources, leaving less for everything else. Whether housing constraints propagate into other lifetime outcomes for younger cohorts, our analysis does not directly address. The mechanism is clear, however: a housing environment that prices out younger cohort members from ownership and burdens renters with high ongoing costs leaves fewer degrees of freedom for the other choices that may shape a cohort's life trajectory.

III. Model

We build a finite-horizon lifecycle model of housing tenure and quality choice with cohort-specific earnings, house prices, interest rates, and mortality risk. Households make discrete decisions over homeownership and housing quality each period, and continuous decisions over consumption and savings. Idiosyncratic uncertainty arises from persistent earnings shocks and stochastic mortality; agents have perfect foresight over house prices, interest rates, and survival probabilities. The model is solved by backward induction using the DC-EGM algorithm (Iskhakov et al., 2017).[5]

before falling by more than a third through 2024 as financing and construction costs rose (U.S. Census Bureau, New Residential Construction).

[5]Details of the solution method, including the upper envelope algorithm and grid construction, are provided in Appendix G.

III.A. Demographics and Timing

An agent enters the model at age 25 and lives to at most age $\bar{a}$, which varies by birth cohort.[6] Retirement occurs at age 65. Time is discrete and indexed by $t = 1, \ldots, T$, where $t = 1$ corresponds to age 25.[7] The model is solved independently for each cohort: cohorts share the same preference structure but face data-based distinct earnings profiles, house prices and interest rate paths, initial conditions, and survival probabilities that depend on birth year.

Between periods t and $t{+}1$, the agent survives with cohort-specific probability π_t, constructed from the Social Security Administration's cohort life tables as a population-weighted unisex average of male and female survival rates.[8] Upon death, all remaining wealth is bequeathed. At $t = T$, $\pi_T = 0$ by convention.

III.B. Bellman Equation

The state at the beginning of period t is

$$(a_t, h_t, q_t, s_t)$$

: liquid assets (a_t, negative if mortgaged), tenure (h_t, 0 = renter, 1 = owner), housing quality (q_t), and the persistent earnings shock (s_t). The variables h_t and q_t reflect the housing arrangement inherited from the previous period.

Within period t, the agent chooses new tenure h' and quality q', which determine the housing services consumed in the current period. The agent also chooses consumption c_t and end-of-period savings a_{t+1}. The current-period choice (h', q') becomes the beginning-of-period state (h_{t+1}, q_{t+1}) in the next period.

The problem has a two-stage structure. In the first stage, conditional on tenure choice h', the agent chooses the optimal quality and continuous allocations:

$$\tilde{V}_t(a_t, h_t, q_t, s_t;\, h') = \max_{q'} \left\{ \max_{c_t,\, a_{t+1}} \Big[u(c_t,\, h',\, q') + \beta\, \pi_t\, \mathbb{E}_{s'|s_t}\big[V_{t+1}(a_{t+1}, h', q', s_{t+1})\big] \right.$$
$$\left. +\ \beta(1-\pi_t)\, \phi_b\, B\big(b(a_{t+1}, h', q')\big) \Big] - \Theta_t \right\} \quad (1)$$

[6] $\bar{a} = 90$ for the 1940 cohort, 92 for 1960, and 94 for 1980. This is a terminal horizon, not an expected lifespan: mortality operates through the cohort survival schedule π_t at every age, and $\bar{a}$ is set beyond the ages that shape the moments we target. It rises across cohorts with observed longevity.

[7] $T = \bar{a} - 25 + 1$.

[8] See Appendix B for the construction of the cohort survival schedules.

subject to the budget constraint (2) and the borrowing constraints below.

The agent enters period t with state (a_t, h_t, q_t, s_t), receives income y_t, and chooses (h', q', c_t, a_{t+1}). The chosen housing arrangement (h', q') provides services in the current period. Housing transitions follow a liquidate-and-rebalance convention: the existing housing position is settled before the new choice is implemented.

$$c_t + a_{t+1} = \underbrace{a_t\big(1+r(a_t)\big) + y_t}_{\text{financial resources}} + \underbrace{h_t \cdot P_t \cdot q_t}_{\substack{\text{housing equity} \\ \text{released}}} - \underbrace{h' \cdot P_t \cdot q'}_{\substack{\text{housing equity} \\ \text{committed}}} - F_t - TC_t \tag{2}$$

When the agent rents (h'=0), the housing commitment term vanishes. When the agent did not previously own (h_t=0), no equity is released. F_t is housing related cash flows (11) and TC_t is transaction costs involved in any buying and selling of homes (12).

In the second stage, the agent observes i.i.d. Type-I Extreme Value taste shocks (ϵ_0, ϵ_1), one for each tenure option, and selects. Let's indicate $x_t = (a_t, h_t, q_t, s_t)$:

$$h'^{*} = \arg\max_{h' \in \{0,1\}} \Big\{ \tilde{V}_t(x_t;\, h') + \sigma_\lambda\, \epsilon_{h'} \Big\} \tag{3}$$

The ex-ante value function, before the taste shocks are realised, takes the closed-form logsum:

$$V_t(x_t) = \sigma_\lambda \ln\left(\exp \frac{\tilde{V}_t(x_t;\, 0)}{\sigma_\lambda} + \exp \frac{\tilde{V}_t(x_t;\, 1)}{\sigma_\lambda} \right) \tag{4}$$

and the implied ownership probability is:

$$\Pr(h' = 1 \mid x_t) = \frac{\exp\big(\tilde{V}_t(x_t;\, 1)/\sigma_\lambda\big)}{\exp\big(\tilde{V}_t(x_t;\, 0)/\sigma_\lambda\big) + \exp\big(\tilde{V}_t(x_t;\, 1)/\sigma_\lambda\big)} \tag{5}$$

The scale parameter σ_λ governs the variance of taste shocks. As $\sigma_\lambda \to 0$, the model converges to a deterministic discrete choice; larger values smooth the tenure decision and capture unobserved heterogeneity in tenure preferences (job mobility, family circumstances, or cultural attitudes toward homeownership) that are not explicitly modelled.

III.C. Preferences

The per-period utility function is separable over non-housing consumption c, housing quality q', and homeownership status h':

$$u(c,\, h',\, q') \;=\; \frac{c^{1-\gamma} - 1}{1-\gamma} \;+\; \phi_q\, \frac{g(q')^{\,1-\psi} - 1}{1-\psi} \;+\; \nu \cdot \mathbb{1}\{h' = 1\} \tag{6}$$

where γ is the coefficient of relative risk aversion, ψ governs the curvature of housing quality utility, ϕ_q is the weight on housing quality, and ν captures the non-pecuniary benefit of ownership. Housing enters twice and differently: q' scales the budget one-for-one, while $g(q')$ is an index of the housing services a tier delivers, so services rise with the size of the home but not in proportion to its value. Appendix D sets out g.[9]

Upon death, the agent's net worth is bequeathed. Liquid savings earn one period of interest before transfer:

$$b(a_{t+1}, h', q') \;=\; a_{t+1}\big(1 + r(a_{t+1})\big) \;+\; h' \cdot P_{t+1} \cdot q' \tag{7}$$

where $r(\cdot)$ is the rate defined in (14) and P_{t+1} is the next-period house price. The bequest utility function is:

$$B(b) \;=\; \frac{b^{1-\omega} - 1}{1 - \omega} \tag{8}$$

with curvature ω and weight ϕ_b in the value function. The bequest motive is of the warm-glow type: the agent derives utility from the amount left behind, but the transfer is not received by any modelled agent, and there is no explicit intergenerational linkage across cohorts. Any such linkage enters only through the age-25 wealth distribution, which is drawn from data in which some transfers have already taken place; transfers received after age 25 are outside the model.

III.D. Income Process

Working period Log gross earnings follow a cohort-specific deterministic age profile with a persistent shock:

$$\ln y_{it}^{\text{pre}} = \alpha_0 + \alpha_1 \cdot \text{age}_t + \alpha_2 \cdot \text{age}_t^2 + s_{it}, \qquad s_{it} = \rho\, s_{i,t-1} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0,\, \sigma_{\varepsilon,t}^2) \tag{9}$$

where ρ is the persistence of the shock. Here i indexes households and t the model period, with $t = 1$ at age 25, so that age is a deterministic function of the period, $\text{age}_t = 24 + t$. Age therefore carries no household subscript: within a cohort every household is the same age at t, and households differ only through the realisation of s_{it}. We solve the model one cohort at a time and suppress the cohort index: the age coefficients and the level of the innovation standard deviation are set separately for each cohort, while ρ and the age profile of $\sigma_{\varepsilon,t}$ are common to all three. The three age coefficients are exactly identified, cohort by cohort, by median real earnings at ages 30, 50 and 60 in the PSID, and the entry

[9] We allow γ, ψ, and ν to differ between working life and retirement (indexed by 1 and 2, respectively), but set $\gamma_1 = \gamma_2$ in the baseline calibration.

value $\sigma_{\varepsilon,1}$ is recovered from the cross-sectional dispersion of log earnings at age 30 through the stationary relation $\text{Var}(s) = \sigma_{\varepsilon,1}^2/(1-\rho^2)$. The innovation standard deviation then widens over the working life, so that the cross-sectional spread of earnings rises with age as it does in the data. Appendix C gives ρ, the cohort-specific coefficients, the profile of $\sigma_{\varepsilon,t}$ and the discretisation.

After-tax income follows the HSV functional form (Heathcote et al., 2017):

$$y_t = \lambda \cdot (y_t^{\text{pre}})^{1-\tau_w} \tag{10}$$

where λ is a scale parameter and τ_w governs progressivity during working life. We set $\tau_w = 0.18$, closely following the HSV estimate of $\tau^{US} = 0.181$ obtained from PSID data (Heathcote et al., 2017, Figure I).

The HSV tax function is applied to earnings, defined as income excluding capital gains and interest income. Our calibration targets for the deterministic earnings profile are constructed from PSID total family income, adjusted by a factor of 0.92 to obtain an earnings measure. This factor corresponds to the average ratio of family earnings to total family income observed in the PSID over 2005–2023, a period for which detailed earnings components are continuously available. The ratio is stable over this period.

The scale parameter λ is set to 1.5, which, combined with $\tau_w = 0.18$, delivers effective average tax rates of approximately 19% at \$30k, 26% at \$50k, and 35% at \$100k of gross earnings.[10]

Retirement Retirement income is determined at the last working period and held constant thereafter. It consists of Social Security benefits, computed from a piecewise-linear Primary Insurance Amount formula, and private pension income modelled as an income-dependent savings rate applied to final earnings.[11] Retirement income is taxed at progressivity $\tau_r \leq \tau_w$.

III.E. Housing

Tenure and quality Each period, the agent chooses tenure $h' \in \{0, 1\}$ (rent or own) and quality $q' \in \{q_1, q_2, q_3\}$, corresponding to small, medium, and large homes. Tiers are defined by room count in the PSID, and q_j is the median house value of tier j relative

[10] These rates are broadly consistent with the federal average tax rates (individual income plus payroll) that the Congressional Budget Office reports rising with income (Congressional Budget Office, 2024). See Appendix F for a full schedule of implied effective average tax rates across the income distribution.

[11] Details of the retirement income formula, its calibration, and the Social Security parameters by cohort are in Appendix E.

to tier 1, so that $q = (1, 2, 3.1)$. A home of quality q' has market value $P_t q'$, and rent, maintenance and property tax all scale with q'. The same three values are used for every cohort; only P_t moves.

House prices The unit house price P_t follows a cohort-specific deterministic path. The historical segment (1960–2024) is constructed from the Median Sales Price of Houses Sold for the United States (FRED series MSPUS), deflated to real terms using the CPI. For the projection period (2025 onward), the baseline path continues at a constant 1.0% annual real growth rate, the post-war long-run average, and this baseline is used throughout the main results. Alternative projection paths, and their effect on the 1980 cohort's late-life outcomes, are examined in Section VI. A home of quality q has market value $P_t \cdot q$.

Flow housing costs

$$F_t = \begin{cases} r_R \cdot P_t \cdot q' - S_t & \text{if } h' = 0 \ \text{ (rent net of subsidy)} \\ (\kappa + \tau_p) \cdot P_t \cdot q' & \text{if } h' = 1 \ \text{ (maintenance + property tax)} \end{cases} \tag{11}$$

where r_R is the rent-to-price ratio, κ the maintenance rate, τ_p the property tax rate, and S_t the rent subsidy described in Section III.G.

Transaction costs Buying a home incurs a proportional cost τ_b and selling incurs τ_s:

$$TC_t = \underbrace{\mathbb{1}\{h_t{=}1 \text{ and } (h'{=}0 \text{ or } q' {\neq} q_t)\}}_{\text{selling current home}} \cdot \tau_s \, P_t \, q_t + \underbrace{\mathbb{1}\{h'{=}1 \text{ and } (h_t{=}0 \text{ or } q' {\neq} q_t)\}}_{\text{purchasing new home}} \cdot \tau_b \, P_t \, q' \tag{12}$$

Owners remaining in the same-quality home ($h_t = h' = 1$ and $q' = q_t$) incur no transaction cost.

Switching cost Any change in tenure or quality incurs a utility cost:

$$\Theta_t = \theta \cdot \mathbb{1}\{h_t \neq h' \text{ or } q_t \neq q'\} \tag{13}$$

where θ may differ between working life (θ_1) and retirement (θ_2). This cost is distinct from the monetary transaction costs TC_t. A monetary cost (say, a brokerage fee proportional to

house value) reduces cash available for consumption. Under CRRA preferences, this hits lower-income households harder: a $10,000 fee hurts a household earning $30,000 far more than one earning $100,000. The switching cost θ, by contrast, is levied directly in utility units and therefore represents the same welfare loss regardless of income or wealth.

Together, TC_t and Θ_t capture the financial burden of moving (which falls disproportionately on less affluent households) and the non-pecuniary disruption of search, relocation, and attachment to a home.

III.F. Financial Markets

Interest rates The return on savings and the cost of borrowing differ:

$$r(a) = \begin{cases} r_s(t) & \text{if } a \geq 0 \quad \text{(saving rate)} \\ r_m(t) & \text{if } a < 0 \quad \text{(mortgage rate)} \end{cases} \tag{14}$$

Both rates are time-varying and cohort-specific. The saving rate is the 1-year Treasury yield deflated by realised CPI inflation; the mortgage rate is the 30-year fixed rate (FRED series MORTGAGE30US) similarly deflated. For years without mortgage data (pre-1972) and for the projection period (post-2024), we use constant rates of 1.5% and 3.5%, comparable to those in Cocco (2005) and Attanasio et al. (2012). Figure 4 plots the two series and the implied spread.

Borrowing Three constraints govern how much a household can borrow. First, end-of-period debt is capped by an unsecured limit, available to renters and owners alike, plus a loan-to-value limit that applies only against a home the household owns:

$$a_{t+1} \;\geq\; -\underbrace{\underline{c}\, y_t \mathbb{1}\{t < t_{\text{ret}}\}}_{\text{unsecured}} \;-\; \underbrace{\text{LTV} \cdot P_t \cdot q' \cdot \mathbb{1}\{h' = 1\}}_{\text{mortgage}} \tag{15}$$

with $\underline{c} = 0.10$, so the unsecured limit is 10% of current income and lapses at retirement. Debt beyond that limit is a mortgage balance, which only owners can hold: a renter's borrowing is the unsecured term alone.

Second, first-time buyers ($h_t{=}0,\ h'{=}1$) must fund the equity share from pre-existing liquid wealth, excluding current-period income:

$$\max(0,\ a_t) \;\geq\; (1 - \text{LTV}) \cdot P_t \cdot q' \tag{16}$$

Third, a debt-to-income ceiling applies at purchase whenever a mortgage is required. Let

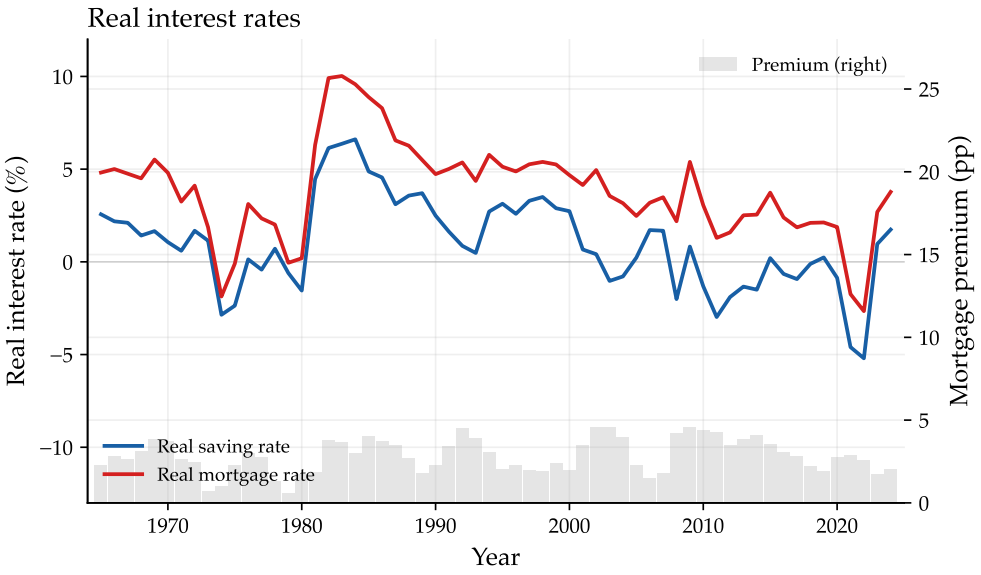


Figure 4. Real Interest Rates and Mortgage Premium, 1965–2024

Notes: Real saving rate = 1-year Treasury yield minus CPI inflation. Real mortgage rate = 30-year fixed rate minus CPI inflation (FRED series MORTGAGE30US). Premium = mortgage rate minus saving rate.

$M_t = \max\big(0,\ P_t\, q' - \max(0, a_t)\big)$ denote the mortgage amount. When $M_t > 0$:

$$\frac{\text{AnnualPayment}\big(M_t,\ r_m,\ \min(30,\ T-t)\big)\ +\ \tau_p \cdot P_t \cdot q'}{y_t}\ \leq\ \bar{d} \tag{17}$$

where the annual payment is computed from a standard fixed-rate amortization formula over $\min(30,\ T-t)$ years, and $\bar{d}$ is the maximum debt-to-income ratio.[12] If the agent can purchase outright ($M_t = 0$), the DTI constraint does not bind.[13]

III.G. Rent Subsidy

Income-eligible renters may receive a housing subsidy modelled on the U.S. Section 8 Housing Choice Voucher program. A renter with gross income $y_t^{\text{pre}} \leq y_t$ occupying a

[12]The numerator includes principal, interest, and property taxes only. It excludes the holding cost κ, which bundles insurance, maintenance, and other charges. The resulting DTI constraint is more lenient than the standard PITI-based underwriting threshold.

[13]The model applies no rate- or credit-based screening beyond the down-payment, LTV, and DTI constraints: any household that clears them borrows at the same rate. In practice lenders denied 9.4 percent of home-purchase applications in 2023, and debt-to-income was among the reasons given for roughly 40 percent of those denials (Liu et al., 2024). Lenders also price borrower risk into the rate, so the model likely overstates the pool of marginal buyers and the rate-sensitivity of young homeownership.

qualifying unit ($q' \leq \bar{q}$) receives:

$$S_t = \max\big(0, \;\; r_R \cdot P_t \cdot q^{\text{FMR}} - \delta \cdot y_t\big) \tag{18}$$

where δ is the tenant burden cap (set to 30% of post-tax income, following HUD guidelines), q^{FMR} is the Fair Market Rent quality benchmark, and $\bar{y}$ is the gross income eligibility threshold.

In our calibration, q^{FMR} is set to q_1 (small homes) and eligibility extends to tenants occupying units of quality up to q_2 (medium). A household renting a medium-quality unit may receive a subsidy, but the maximum subsidy is capped at the small-home rent level; any rent in excess of the 30% burden threshold must be paid out of pocket. Households renting large homes (q_3) are ineligible for the program.

IV. Calibration

We calibrate the model to the 1940 cohort by the method of simulated moments and hold the estimated parameters fixed. The 1960 and 1980 cohorts are then obtained by re-solving the model with only the exogenous inputs changed: house prices, interest rates, and earnings. Survival schedules and the age-25 distribution of income, wealth, and tenure are also cohort-specific, set from the data for each cohort rather than estimated.

IV.A. Estimation

One set of parameters is fixed before estimation (Table 2, Panel A). The earnings process is a log AR(1) with age-varying innovation variance, estimated from the PSID and detailed in Appendix C. Interest and mortgage rates follow the realised series, and taxes use the HSV schedule (Heathcote et al., 2017), with lower progressivity in retirement to reflect the treatment of Social Security. We fix risk aversion at $\gamma = 1.6$: the moments we target do not pin it down, and this value is within standard ranges in the lifecycle literature. The bequest curvature is tied to γ.

The remaining nine parameters (Panel B) are estimated jointly by the method of simulated moments against thirteen moments from the PSID (Table 4). They minimise

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_m w_m \left[\frac{m^{\text{mod}}(\boldsymbol{\theta}) - m^{\text{dat}}}{m^{\text{dat}}} \right]^2, \tag{19}$$

with weights w_m inversely proportional to each moment's squared coefficient of variation, so more precisely measured moments count for more. The tenure-transition rates, the

wealth levels and the share of households still solvent at 75 carry a larger effective standard error, reflecting that the model abstracts from the forces that move them (relocation and life-event shocks, and medical and longevity risk; Section IV.B), so their remaining gap reflects what the model leaves out rather than the values we chose for the parameters. The homeownership and quality-tier shares, which the model is built to explain, are measured precisely and carry the greatest weight. Minimization uses differential evolution, followed by a Nelder–Mead simplex around its best point.

Although the nine parameters are estimated jointly, each moves a distinct set of moments most sharply, and it helps to see which (Table 2, last column). The discount factor β is identified by wealth accumulation over working life: a more patient household saves more, so the working-age wealth-to-income ratio rises with β. Because the model underaccumulates wealth at every age, that moment is down-weighted, and β is disciplined alongside the ownership and quality moments through the wealth a household needs before it can buy. The bequest weight ϕ_b plays the same role at the other end of life, setting how slowly households draw wealth down, so it is read from how much net worth they carry into old age. The ownership premia ν_1, ν_2 fix how much a household values owning for its own sake and so move the homeownership rate up or down; splitting them between working life and retirement lets the two ownership levels be matched separately. The switching costs θ_1, θ_2 work in the opposite direction, taxing any move, so they are pinned by the tenure-transition rates, again split by phase. The cost applies to quality changes as well, but we do not target quality-transition rates: the model has three tiers, which is a coarse approximation of the true quality choice, and much observed movement between them follows relocation and life events the model does not contain. The quality weight ϕ_q sets the overall pull toward better housing, and hence the share of households in the top tier, while the curvatures ψ_1, ψ_2 let that share differ between working life and retirement, disciplining the shape of the quality profile over the lifecycle and not only its level.

Two preference parameters bear directly on the decision to buy: the ownership premium ν, which a household collects in every year it owns, and the switching cost θ, which it pays whenever it changes tenure or quality. Everything else in the problem reaches the tenure choice through prices and income. Table 3 states both in dollars, as the consumption a household would trade for them. Both are larger the higher a household's level of consumption, since a given utility gain takes more consumption to replace when consumption is already high. The closing cost on a purchase is a fixed sum and does not grow. That difference changes which of the two costs binds as we move up the distribution. At the bottom the financial cost is the larger barrier, \$39k in closing fees against \$18k for the non-monetary cost of moving. The two are about equal around the median. At \$100k of

consumption the fee has become minor and the non-monetary cost, at \$97k, is two and a half times larger. That the estimated moving cost lands in the same range as real closing fees is some reassurance that the model is not buying its mobility fit with an implausible number.

Set against these one-time costs, the ownership premium is small in any single year, \$4.5k at the bottom and \$45k at the top. What makes buying worthwhile is that the premium arrives every year the household stays, while the costs are paid once. The table therefore reports what the premium is worth after ten years of ownership, and the tenure at which it finally covers the cost of entry. This is where the two preferences interact. A household at \$100k of consumption recovers its entry cost in four years, and one at the median in eight. A household at \$20k needs twenty-three. Ten years of ownership leaves it \$22k short, because the premium it earns is small in proportion to a closing fee that is the same for everyone.

The asymmetry is not a statement about impatience, since all three face the same discount factor. It follows from a fixed dollar cost meeting a benefit that scales with living standards. A household that spends more gets more out of owning in each year it owns, while paying the same closing fee as everyone else, so it recovers that fee sooner. The households that would wait longest to recover it are also the ones least able to pay it in the first place. The calculation counts only the non-pecuniary premium and sets the house aside as an asset, so it understates the return to buying at every level. The ordering across consumption levels is what matters here.

Households enter at age 25 with a joint distribution of income, tenure, quality, and wealth drawn from the PSID. The three cohorts differ only here and in their exogenous paths. Income is drawn from the stationary distribution of the earnings process; tenure from PSID ownership rates by income decile; quality from PSID quality shares, separately for owners and renters; and wealth to match the PSID quartiles by tenure. The wealth draws are then assigned across households so that higher-income households tend to receive larger draws. Because the assignment only reshuffles a fixed set of draws, the wealth distribution is exactly the fitted one; only its association with income changes.[14] The 1940 cohort starts with a 32% ownership rate and positive median wealth for both owners and renters; by 1980 the entry ownership rate falls to 17%, median renter wealth turns negative, and the income–ownership gradient flattens. These differences, together

[14]Two floors are applied afterwards, the loan-to-value limit for owners and the unsecured borrowing limit for renters. The realised 25th percentile therefore sits above its target (owners \$24.8k against \$15.1k; renters $-$\$1.4k against $-$\$8.9k), while the median and 75th percentile are matched. The truncation removes the most fragile entrants, so the welfare losses we report for low-wealth households in Section V.C.3 are, if anything, understated.

with the divergent price and rate paths each cohort faces, are what the model translates into lifecycle outcomes.

Table 2. Parameters

Parameter	Description	Value	Identified by / source
A. Set externally			
γ	Risk aversion	1.60	Literature; weakly identified
λ	After-tax income scale (HSV)	1.50	HSV
τ_w, τ_r	Tax progressivity (work, ret.)	0.18, 0.10	HSV / Social Security
κ	Maintenance (% of house value)	0.015	Literature
τ_p	Property tax rate	0.015	U.S. effective rate
τ_b	Buying transaction cost	0.05	Literature
τ_s	Selling transaction cost	0.06	Literature
r_R	Rent-to-price ratio	0.055	Literature
$r,\ r_m$	Interest & mortgage rates	time-varying	FRED
LTV	Maximum loan-to-value	0.90	Institutional
DTI	Debt-to-income cap	0.43	Institutional
B. Estimated (MSM)			
β	Discount factor	0.941	Wealth/income, 25–64
ϕ_q	Quality weight	1.914	Level of the top-tier quality shares
ψ_1, ψ_2	Quality curvature (work, ret.)	7.132, 7.439	Working–retirement gap in those shares
ν_1, ν_2	Ownership premium (work, ret.)	0.046, 0.034	Homeownership by age band
θ_1, θ_2	Switching cost (work, ret.)	0.802, 0.649	Tenure transition rates
ϕ_b	Bequest weight	0.642	Late-life wealth: net worth at 80, positive net worth at 75, wealth/income 65+

Notes: λ and τ_w, τ_r are the scale and progressivity of the HSV after-tax schedule $y^{\text{net}} = \lambda\, y^{1-\tau}$. The earnings process (persistence and innovation variance) is reported in Appendix C. The last column names moments reported in Table 4. All nine parameters are estimated jointly against all thirteen moments; the column names the moments each parameter moves most sharply, not an exclusive assignment.

IV.B. Targeted Moments

Table 4 reports the targeted moments against the model at the estimated parameters. The model matches the homeownership lifecycle closely: it reproduces the rapid entry into

Table 3. What the Estimated Preferences Are Worth, in Consumption

Consumption	One-time cost		Ownership premium ν		Years to
at age 25	buying (tier-2)	switching θ	per year	held 10 yrs	break even
Low ($20k)	38.6	17.9	4.5	35	23
Median ($40k)	38.6	37.6	12.4	95	8
High ($100k)	38.6	97.2	45.1	347	4

Notes: All entries in thousands of 2022 dollars, computed from the calibrated parameters. For a household consuming c, the consumption-equivalent of a fixed utility amount X is the share of consumption it would give up to obtain it, $\tau_X(c) = 1 - [1 - X(1-\gamma)/c^{1-\gamma}]^{1/(1-\gamma)}$, in dollars $c\,\tau_X(c)$; here $X = \nu_1$ (ownership) or θ_1 (switching). Ownership is a flow received in every year of tenure, so the held-T column accumulates it with discounting and survival from age 25, and break-even is the shortest tenure at which that sum covers both one-time costs:

$$\mathrm{PV}_T(c) = \sum_{t=0}^{T-1} \beta^t S_t \; c\,\tau_{\nu(t)}(c), \qquad T^\star(c) = \min\{\, T : \mathrm{PV}_T(c) \;\geq\; \text{buying cost} + \text{switching cost}(c) \,\}.$$

Here S_t is survival to t years past 25 and $\nu(t)$ is ν_1 before age 65 and ν_2 after, so the annuity factor $\sum \beta^t S_t$ is 7.7 at ten years against 15.9 over the whole horizon: at $\beta = 0.94$ it is discounting rather than mortality that limits it. Break-even counts only the non-pecuniary premium and excludes capital gains and the resale value of the house, so it is a lower bound on the return to buying. The two costs are one-time and are not capitalized. The buying cost is the model's closing cost τ_b on a tier-2 house at its 2024 price ($771k, in 2022 dollars; 5%), and does not vary with consumption. The three consumption levels are roughly the 10th, 50th, and 90th percentiles of the CEX distribution of household consumption at age 25.

ownership over the working years and the near-flat profile in retirement, which is what the paper is built to explain. The share of households in the top quality tier is also close to its target.

The model matches two sets of targeted moments only loosely. The first is tenure transitions, where both rates come in below target and the own-to-rent rate in retirement is the further off. Owners in the model rarely sell once they have bought. Both rates are low on either side, under two percentage points, so the gap is a matter of how rare these moves are rather than whether they happen at all. Housing turns over infrequently in any case, whether measured in the data or implied by models with realistic transaction costs (Flavin and Yamashita, 2002; Bajari et al., 2013). The model has no relocation or life-event shocks to prompt a move, and its housing choice is coarse, with three quality tiers and no location dimension, so households have few margins along which to adjust. The model also under-accumulates wealth: the working-age wealth-to-income ratio and median net worth at age 80 both fall short of the data. That gap has a separate source: the model carries no medical-expense, so the old save too little for precaution. We keep both sets of moments in the estimation, where they still discipline the ownership premia and the discount factor, but we read the fit through the homeownership and quality dimensions the model is built

to match.

Table 4. Targeted Moments

Moment	Data	Model	Distance
Homeownership, 25–34	0.534	0.543	+0.009
Homeownership, 35–44	0.726	0.719	−0.007
Homeownership, 45–54	0.786	0.790	+0.004
Homeownership, 55–64	0.807	0.809	+0.002
Homeownership, 65–74	0.812	0.817	+0.005
Wealth/income, 25–64	2.50	1.49	−1.01
Wealth/income, 65+	5.73	5.86	+0.13
Top-tier quality, 35–44	0.414	0.415	+0.001
Top-tier quality, 65–74	0.381	0.385	+0.004
Rent-to-own rate, working	0.031	0.013	−0.018
Own-to-rent rate, retirement	0.007	0.001	−0.007
Median net worth at 80 ($k)	340	217	−123
Positive net worth at 75	0.946	0.999	+0.053

Notes: Targets are computed from the PSID for the 1940 cohort. Distance is model minus data, in the units of the moment. Moment numbers are cited in Table 2.

IV.C. Untargeted Moments

The estimation uses only the 1940 cohort, so every moment here is untargeted, and the 1960 and 1980 cohorts are entirely out of sample: we re-solve the model for them with new exogenous paths and initial conditions but the same estimated parameters. The cross-cohort homeownership decline that follows is the paper's main object, and we take it up in Section V.A rather than here.

The test we report here fixes the age band and reads across the income distribution. Figure 5 shows homeownership by income decile at ages 35–44, the band where all three cohorts are observed, with the data on the left and the model on the right. Only the 1940 band average is targeted, so the slope across deciles and both younger cohorts are free. The model gets the shape right in both directions: ownership rises with income within every cohort, and at each decile it steps down from 1940 to 1960 to 1980, with band averages within three points of the data. Two gaps are worth naming. The model's income slope is a little too flat, putting more of the bottom decile into ownership than the data does, clearest for the 1940 cohort; and at the top decile the cohorts nearly meet in the model while the data keep them apart.

The model likewise captures the level and mid-life hump of non-housing consumption

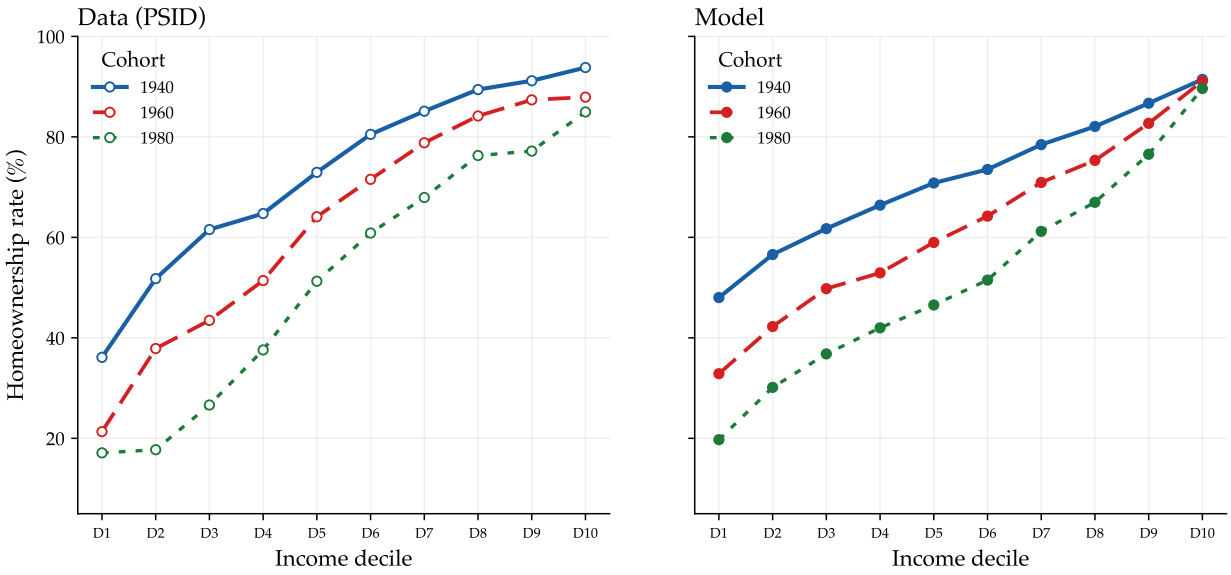


Figure 5. Homeownership by Income Decile, Ages 35–44 (Untargeted)
Notes: Homeownership rate by within-band income decile at ages 35–44, the band at which all three birth cohorts are observed in the PSID. Data on the left, model on the right; each line is a cohort. Deciles are formed within cohort and survey year.

in the CEX, a dimension absent from the criterion.[15]

The model is less accurate on two counts. It under-accumulates liquid wealth and holds most retirement resources as home equity, so financial assets run down to near zero and old-age net worth is too compressed across income groups; and consumption is less dispersed than in the data. Both follow from the same feature already visible in the targeted fit: thin liquid savings alongside heavy housing wealth.

V. Results

We organise the model's output around three findings about how the 1980 cohort fares relative to the 1940 baseline.

First, younger cohorts own less, and the homes they own are smaller. The entry into homeownership shifts later across generations, and conditional on owning, households sort into lower quality tiers. A Shapley decomposition traces the bulk of the gap to house prices.

Second, non-housing consumption is compressed at younger ages. The compression is concentrated in the entry window, where housing absorbs a larger share of income for

[15]Non-housing consumption (CEX, 1940 cohort; $k, median over all households) at ages 25/35/45/55/65 is 38.3/50.0/52.3/42.3/38.9 in the data against 35.5/42.1/47.3/48.0/42.8 in the model. The model's peak is flatter and arrives about a decade later.

each successive cohort, and persists within both renter and owner cells.

Third, the welfare loss from the 1940-to-1980 environmental shift is substantial and unequally distributed across initial states. We measure it ex ante, from the perspective of an agent at the start of the lifecycle given their earnings state, assets, and housing position. The decomposition shows that prices and income each contribute materially, with the relative shares varying by initial conditions.

V.A. Younger Cohorts Own Less and Live in Lower-Quality Homes

Both the data and the model show that younger cohorts enter homeownership later, reach lower peak ownership rates, and live in lower-quality homes. A Shapley decomposition attributes most of the gap to house prices at every age, with the earnings contribution growing over the lifecycle.

1. The Tenure Margin

Homeownership rises with age in every cohort, but each successive cohort starts lower and stays lower. Figure 6 plots the lifecycle profile in the data and the model. In the data, the ownership rate in the 25–34 band falls from 53% for the 1940 cohort to 38% for the 1960 cohort and 31% for the 1980 cohort. The gap does not close with age: the 1940 cohort stays above 80% from the 55–64 band onward, a level the 1960 cohort never reaches within its observation window (67% at 55–64). The model reproduces the entry-band rates (54%, 38%, and 29%) and the cohort ordering closely.

For the younger cohorts, the model projects a decline in ownership late in life, from the 55–64 band onward for the 1960 cohort and from the 45–54 band for the 1980 cohort. These age bands lie outside the PSID window, so the paths are forecasts rather than fitted moments. Late-life saving motives in the model are deliberately sparse: there is no medical-expenditure risk of the kind emphasized by De Nardi et al. (2010), the horizon ends at a known terminal period rather than at an uncertain date of death, and the warm-glow bequest motive is bounded. Once expected bequests reach their target, households consume their remaining wealth, and for most retirees that wealth is the house: the median retired homeowner holds roughly 100% of net worth in the house at ages 65 and older, leaving little scope to smooth consumption out of financial assets alone. Eventually the only margin left is to sell. No such decline appears for the 1940 cohort, whose ownership stays high through retirement: retiring in an era of comparatively low house values, its housing wealth is close to what it intends to bequeath, whereas later cohorts retire holding housing well above their bequest target and draw it down by selling.

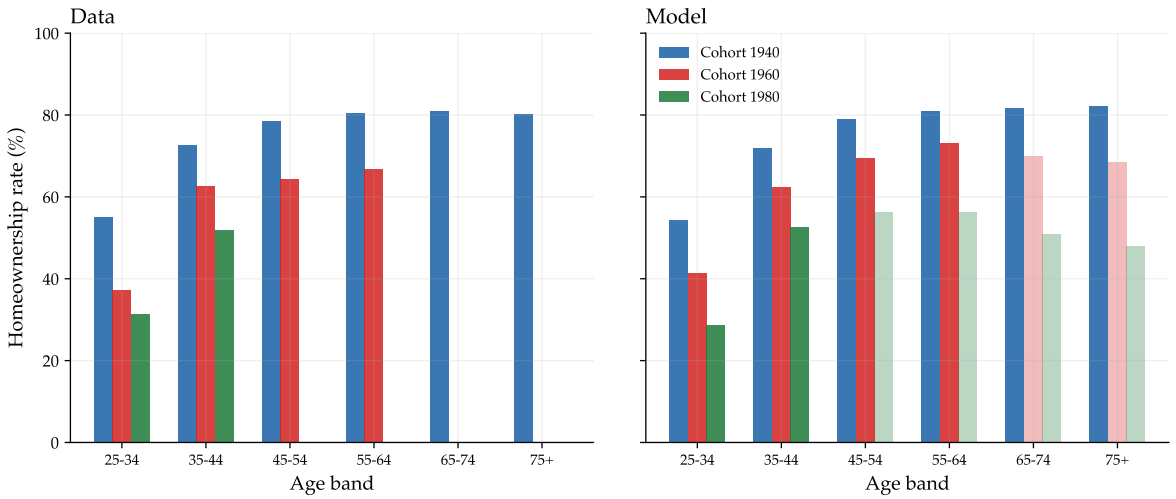


Figure 6. Homeownership Lifecycle by Cohort, Data and Model

Notes: Homeownership rate by birth cohort and age band. Left panel: PSID; right panel: model. In the data panel, bars are shown only for age bands within each cohort's observation window; in the model panel, bars for age bands outside that window (model forecasts) are drawn faded.

These late-life paths are the model's least certain. To turn home equity into consumption, a retiree here can only borrow against the house or sell it, and with borrowing against a small pension unattractive, it sells. Reality lets owners draw on home equity while staying put, and it is the low-income, low-wealth households at issue here who most rely on that option (Nakajima and Telyukova, 2017). Left with only the choice to sell, the model pushes more of old-age adjustment onto tenure than a fuller set of options would. Faced with adjustment costs, households more often leave the housing stock in place and absorb shocks through other spending (Bajari et al., 2013). The projected post-65 decline therefore likely overstates what households would do in practice, and we take it up in the robustness analysis of Section VI.

2. *The Quality Margin*

Ownership rates alone understate the cohort gap, because the quality of housing that households live in, for owners and renters alike, has also shifted down. Among households aged 25–34, the share in the lowest quality tier (Q1) rises from 31% in the 1940 cohort to 44% in the 1960 cohort and 61% in the 1980 cohort in the data. The model produces 37%, 43%, and 65%. The top tier mirrors the shift: the Q3 share at entry falls from 24% to 19% to 14%.

Figure 7 shows that this tilt toward lower tiers persists across the full lifecycle, not only at entry. At middle age, the 1940 cohort recovers Q3 share as earnings and accumulated equity allow households to trade up: from 24% at 25–34 to 46% at 45–54 in the data, a

profile the model tracks closely (27% to 46%). Younger cohorts start lower and remain lower at every band where the cohorts can be compared: at 45–54, the 1960 cohort's Q3 share is 37%, 9 points below the 1940 cohort at the same age.

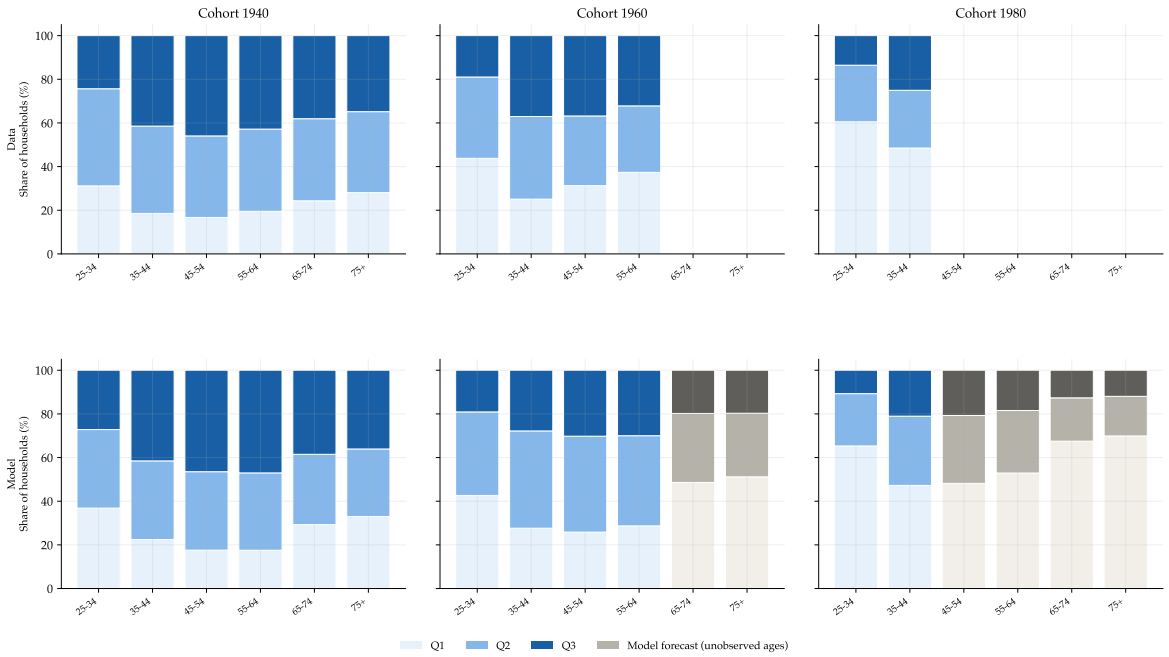


Figure 7. Housing-Quality Distribution by Cohort and Age Band

Notes: Stacked bars show the share of all households (owners and renters) in each housing-quality tier (Q1 = lowest, Q3 = highest) by birth cohort (columns) and age band (x-axis). The top row reports PSID data; the bottom row reports model outcomes. Grey bars mark model forecasts for age bands outside the PSID observation window.

The model reproduces the 1940 cohort's full lifecycle quality profile closely: the climb out of Q1 and into Q3 over the working life emerges from consumption smoothing and optimal trade-up alone, without appeal to socio-demographic change. The cross-cohort ordering holds at every age band, in the model as in the data: younger cohorts live in lower-quality homes over the entire lifecycle. That shift is telling. In the data, part of it may reflect socio-demographic forces, and delayed and less frequent marriage in particular interacts strongly with housing demand. The model contains no such channel. Higher house prices and relatively stagnant earnings early in working life are, on their own, sufficient to generate the pattern.

3. What Drives the Gap

What ownership rate would the 1980 cohort have reached facing the 1940 environment? We swap three channels, house prices, interest rates and the earnings process, from their

1980 to their 1940 values, holding initial conditions at 1980. Because the channels interact, swapping one channel at a time does not split the gap additively. The Shapley decomposition instead averages each channel's marginal contribution over every order in which the three channels can be switched, using all $2^3 = 8$ combinations of 1980 and 1940 values, so the contributions sum to the total by construction.[16]

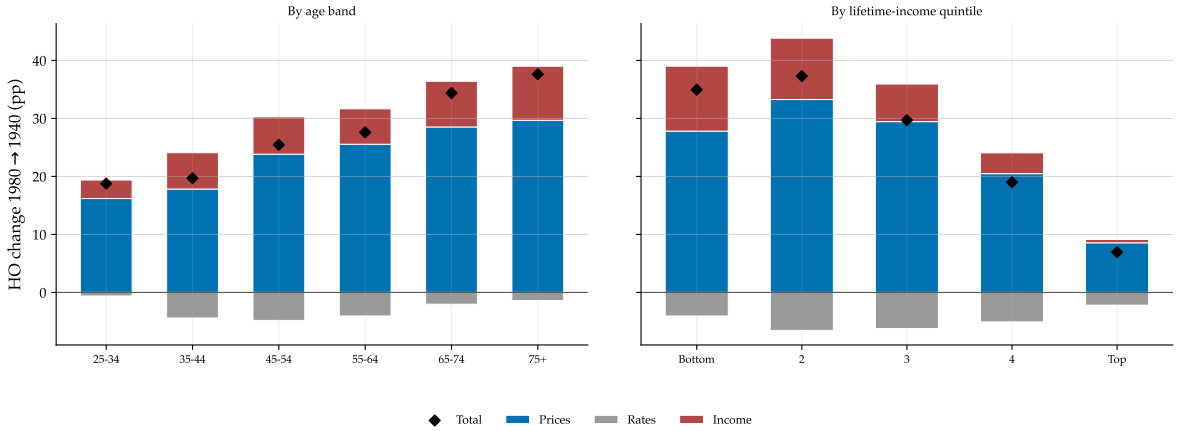


Figure 8. Shapley Decomposition of the 1940–1980 Homeownership Gap

Notes: Bars show the Shapley contribution of each channel, in percentage points, to the counterfactual change in homeownership from moving the 1980 cohort to the 1940 environment: by age band (left panel) and, at ages 45–54, by lifetime-income quintile (right panel).

The 1940 environment sustains higher ownership at every age, and the advantage widens over the lifecycle, from 19 pp at ages 25–34 to 37 pp at 75+, or 27 pp in the aggregate.[17] House prices dominate throughout, and their contribution grows with age: cheaper housing compounds over the tenure history, as its effects on entry, mortgage burden, and trade-up accumulate into wealth. The earnings channel is modest early and rises for a related reason, as the 1980 cohort's earnings deficit accrues into a wealth deficit that binds more tightly the longer it accumulates; at entry both environments leave young households constrained regardless. The right panel repeats the decomposition at ages 45–54 by lifetime income. The gain from the 1940 environment is far larger at the bottom, where ownership rises by about 35 percentage points for the lowest-income fifth against 7 for the highest, and both channels contribute to the gradient. The timing of the gain also differs by income. High earners in the 1980 environment mainly buy later, so at ages 25 to 29 the 1940 environment would raise their ownership by more than that of the bottom

[16] See Shapley (1953) and Shorrocks (2013). Details are in Appendix H.

[17] The split between the earnings and price contributions depends on the income path assigned to the 1980 cohort over ages it has not yet reached. Under an alternative path in which the rise in earnings risk continues, the earnings contribution is larger while the price contribution is essentially unchanged (Section VI), so the age gradient here is a conservative reading of the earnings channel.

fifth, but this gap closes to a few points by their mid-forties, while for the bottom fifth it widens with every age band.[18] On the ownership margin the environment younger cohorts inherited is regressive, withholding ownership most from those least able to afford it.

V.B. The Housing Burden Crowds Out Non-Housing Consumption

Rising housing costs bear on more than the ownership decision: for households that do buy or rent, a growing share of the budget goes to housing, leaving less for everything else. The Consumer Expenditure Survey (CEX) is a repeated cross section, so we use it to read how a fixed age group has fared over calendar time and, where a cohort is observed, to gauge the model. No cross section delivers the counterfactual the argument needs, the same households under a different environment, nor a cohort followed across its whole life, since the survey's span is short. Both are the model's role: it holds preferences fixed, varies only the environment, and traces each cohort over the full lifecycle.

1. Stalled Consumption and a Rising Burden

In the data, the non-housing consumption[19] of the young has been roughly flat in real terms for three decades, In the data, the non-housing consumption of the young has barely grown in real terms for three decades, while that of households aged 65 and over rose (Panel A). The stagnation is specific to the young. The housing share of consumption, meanwhile, climbs from about 0.18 to 0.21 for all households, and among the young the rise is concentrated in renters (0.22 to 0.28), with owners looking flat ($\approx$ 0.20; Panel B). The flat owner cell is tempting to read as owners having escaped the increase, but a more likely account is composition: as first-time purchase narrowed to higher-income households, the marginal young owner is better off, which holds the observed burden down even as the

[18] Gain in homeownership from the 1940 environment, in percentage points, by lifetime-income group fixed in the 1980 environment:

Ages	25–29	30–34	35–44	45–54
Bottom 20%	14.6	23.3	28.0	34.9
Top 20%	22.7	13.9	8.3	6.9
Top 10%	22.0	12.6	6.1	4.1
Top 5%	21.9	10.7	5.0	3.0
Top 1%	24.3	10.8	3.2	0.2

The top 5 and 1 percent contain about 500 and 100 simulated households. The 1980 cohort is observed in the data up to its early forties.

[19] Non-housing consumption is total household spending net of housing and of saving-type outlays. The same age pattern holds for non-durable non-housing consumption, which additionally removes durables. The 2021–23 pandemic spike is omitted from the figure. Including it leaves the differences across age groups unchanged.

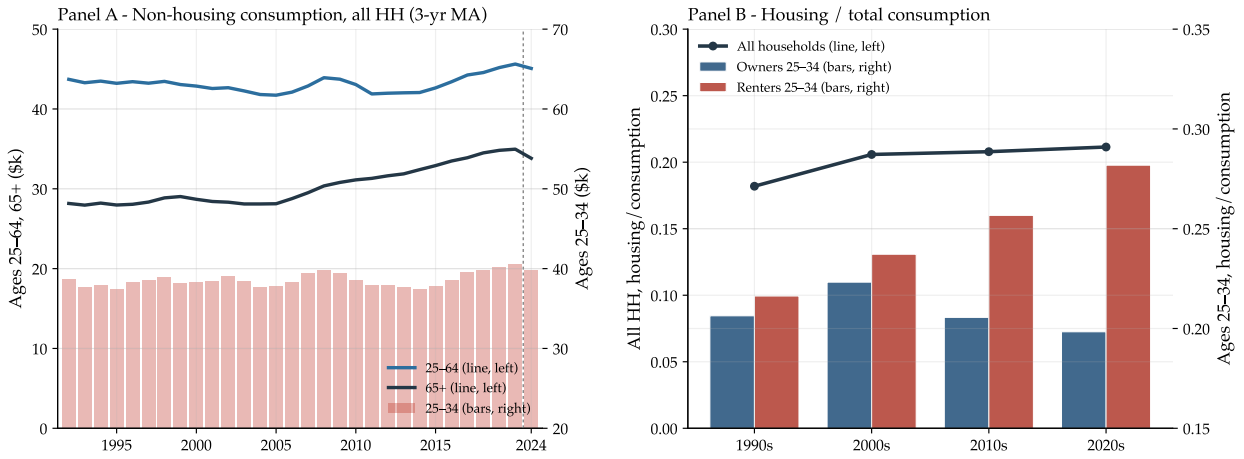


Figure 9. Non-Housing Consumption and Housing Burden, CEX

Notes: CEX, weighted medians, real 2022 dollars. Panel A: all households, three-year moving average; the pandemic years 2021–23 are omitted (marked by the break) and the averages reach across the gap. Panel B: housing / total consumption, median of household ratios (mor); left axis (line) = all households, right axis (bars) = ages 25–34, decade averages. The two axes cover different groups and should be read by slope, not by relative height.

burden on a fixed household rises. The renter series is the cleaner read. Renters are now the large majority of young households, so their composition barely moves as ownership falls, while the shrinking owner group is increasingly selected. Taken together, the natural reading is that young households' non-housing consumption has not grown while their housing burden has.

2. *The Same Squeeze Across Cohorts*

Figure 10 puts both sides of the budget together across cohorts: what housing takes, and what is left for everything else. Every household lives through its own cohort's environment and we observe all of them, so no household drops out of view. Tenure is still chosen, however, so the same composition problem as in the data applies here, and to the same asymmetric degree: at ages 25–34 the median owner's income runs 1.18 times the cohort median in the 1940 cohort but 1.50 times in the 1980 cohort, while for renters the ratio moves only from 0.84 to 0.89. The owner panels compare an increasingly selected group across cohorts; the renter panels do not. The story repeats for owners (Panels A, C) and renters (Panels B, D): non-housing consumption at entry is lower for each later cohort, most of the drop falling between the 1940 and 1960 cohorts and echoing the stagnation we saw in calendar time, while the housing burden at 25–34 moves the opposite way and climbs across cohorts. Since the model conditions only on the cohort's prices, rates, and income process, this is the pure environmental effect.

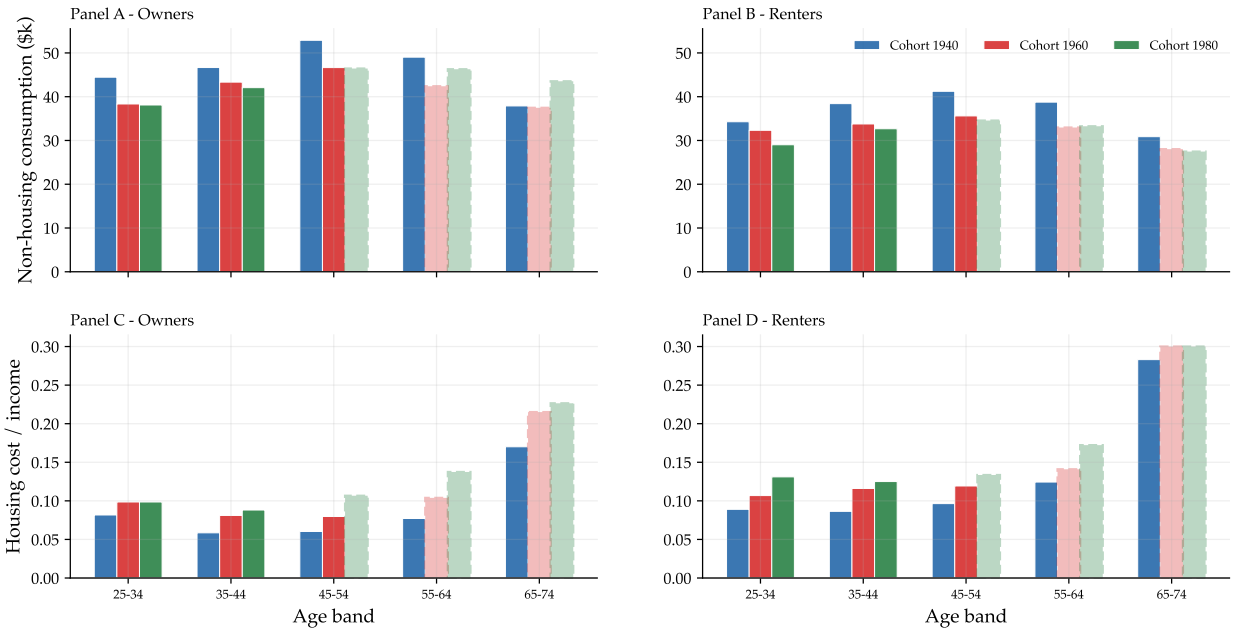


Figure 10. Non-Housing Consumption and Housing Burden by Tenure and Cohort, Model
Notes: Model medians among alive agents, by age band and cohort (1940/1960/1980). Panels A, B: non-housing consumption ($k); Panels C, D: housing cost over post-tax income. Owners in A, C; renters in B, D. Reading a column top to bottom is the squeeze for that tenure. Cohort–ages not yet observed in the data are translucent; the renter burden at 65–74 is bounded by the model's rent-assistance rule (contribution capped at 30% of income, full take-up). Levels sit below the CEX because the model is not calibrated to expenditure levels; the cohort ordering is the comparison of interest.

Two details qualify the picture. Among owners the ordering flips in old age, where later cohorts eventually consume more than the 1940 cohort. This is composition rather than a reversal in living standards. Ownership at 65–74 falls from 82% of the 1940 cohort to 51% of the 1980 cohort, and the 1980 households who still own hold 1.9 times their cohort's median wealth, against 1.2 times for the 1940 cohort. Pooling tenures removes the flip: median consumption at 65–74 runs $36.8k, $34.5k and $35.4k across the three cohorts, and past 75 the 1980 cohort is the lowest of the three. A second qualification runs the other way. The renter burden is held down by a rent-assistance rule (contributions capped at 30% of income, full take-up), whereas the actual Housing Choice Voucher (Section 8) program is rationed rather than an entitlement, reaching only about one in four eligible households (Economy and Finnigan, 2025). Real burdens on low-income renters are therefore higher than the figure shows, and the renter welfare losses in Section V.C are understated to that degree.

Non-housing consumption is the part of the budget a household can freely allocate, so its compression at young ages hits the stage of life when the big commitments are made. The squeeze reaches past the budget into those decisions: leaving home, forming a

household, marriage, and children turn on housing itself, but also on what is left to spend once housing is paid for, and a sustained shortfall could push them later. The evidence points that way. Household formation responds to housing costs directly: Paciorek (2016) finds that once the ageing of the population is stripped out, what remains of the headship rate moves with the cost of housing, and the earlier work he draws on finds that high rents retard the formation of independent households outright. On childbearing, Dettling and Kearney (2014) find that higher house prices lower birth rates among renters and prospective buyers even as they raise them among owners, a tenure split that matches the renter squeeze here. The structural fertility literature places its main levers elsewhere, in the rising opportunity cost of children as wages grow (Manuelli and Seshadri, 2009) and in preferences and intergenerational altruism (Córdoba and Ripoll, 2018). We do not displace these, but add housing cost as one more margin biting in the same years. The channel is asymmetric by tenure: rising house values raise owners' fertility through the wealth effect (Lovenheim and Mumford, 2013) but squeeze renters through the rent burden, the margin our mechanism adds. We model none of them directly, and other forces are surely at work, so we offer the channel as a direction, not a result. It does, though, caution against reading the cohort gap as a purely exogenous "demographics" residual. Demographics matter, and interact with the environment: much of the cross-cohort shift in young households' wealth, for one, tracks changes in their composition (Gale et al., 2020), and a large structural literature runs the causation from demographics to housing, with later marriage and greater income risk delaying entry into ownership (Fisher and Gervais, 2011) and shifting marital transitions moving singles and couples in opposite directions (Chang, 2024). Our budget mechanism suggests the causation can also run the other way, from the housing environment onto the household formation those accounts take as given.

V.C. Ex-ante Welfare: Who Loses, and Why

The ownership and consumption gaps in Sections V.A and V.B are measured in different units (percentage points of tenure, dollars of consumption, ratios of income) and cannot be added. They are nonetheless pieces of one question: is a household in the 1980 cohort better off starting life in the 1940 environment? This section answers it with a single number and traces where that number comes from.

The usual answer to that question is a single aggregate, real consumption, in which housing enters as one deflated component among many. Housing sits awkwardly in that treatment. It is the largest item in most budgets, its relative price has risen over decades, and its incidence is uneven: households that bought early or fixed a mortgage carry a housing cost set at an earlier price, while renters and recent buyers face the current one. A

common deflator applied across households averages over that gap and assigns a uniform cost of housing to cohorts that in fact met very different ones. It also abstracts from quality, so a household paying more for the same or a smaller home registers no loss at all. Whether a later cohort is better off therefore turns on a comparison the aggregate does not surface, between the growth of its earnings and the growth of what housing costs it. The measure below makes that comparison directly, household by household.

1. The Measure

We compare the two environments at age 25, before a household knows how its life will turn out. Write A for the baseline environment (the 1980 world the younger cohort inherited) and B for the alternative, the 1940 environment. Fix a household by where it starts (its age-25 income, assets, tenure, and quality) and let W_i^A and W_i^B be its expected lifetime value under each, each averaged over the age-25 taste shock and every shock after it. The ex-ante compensating variation λ_i is the proportional rise in baseline non-housing consumption that makes environment A as good as environment B:

$$\lambda_i \;=\; \left(1+\tfrac{\Delta W_i}{A_{0,i}}\right)^{\frac{1}{1-\gamma}}-1, \qquad \Delta W_i = W_i^B - W_i^A. \tag{20}$$

The denominator $A_{0,i}=\sum_{t=1}^{T}\beta^{t-1}S_t\,c_{it}^{1-\gamma}/(1-\gamma)$ is the value of the household's baseline consumption stream, weighted by the survival probability S_t so that it prices the consumption a household expects to receive rather than consumption conditional on staying alive. Appendix I derives (20) from the indifference condition. We report λ_i as a percentage: a positive number means environment B (the 1940 world) is preferred, and is the welfare loss carried by the cohort that inherited the 1980 world. Averaging W_i over all shocks and weighting $A_{0,i}$ by survival is what keeps the comparison ex ante, before any shock is realised, so it depends only on the household's age-25 state.[20]

To locate the source of the loss, we roll the environment back to its 1940 values a few features at a time (house prices, interest rates, and the earnings process) and split the total with Shapley values, exactly as in Section V.A.3. Survival probabilities and the terminal horizon are held at their 1980 values throughout.

The compensation is expressed in units of non-housing consumption, and the levels below are therefore larger than the compensating variations reported in lifecycle models

[20] A few households sit where the environment-B gain nearly cancels the whole value of baseline consumption ($A_{0,i}\to 0$), sending (20) to extreme values. They are too few to shape the cohort's experience but enough to swamp a plain mean, so we report the median loss and, as a check, a utility-aggregated loss that averages W and A_0 across households before applying (20) once. Each statistic is decomposed directly, so the pieces still sum to the total.

with a single consumption good. Two things account for the difference, and neither is a matter of presentation.

The first is the unit itself. Housing enters utility here through a discrete quality tier and a tenure indicator, so there is no continuous housing margin to scale. Non-housing consumption is the only good whose quantity can be raised proportionally to restore indifference, and it is around four-fifths of the budget (Section V.B). A single-good model spreads the same adjustment over the whole bundle, which mechanically yields a smaller number for the same underlying welfare change.

The second is that this measure carries a term the other does not. Take a household that owns a medium home in the 1940 environment and rents a small one in the 1980 environment. It pays more for housing, and it also lives in a worse home. A model in which housing is only a cost records the first and stops there, since there is no quality to lose. Here quality is chosen, so the smaller home is a loss in its own right, and the compensation has to cover both.

The levels should be read with the unit in mind. The comparisons across households, which is what the incidence results below rest on, do not depend on it, since every household is measured against its own baseline consumption.

2. The Aggregate Loss

A household that inherits the 1980 environment would need about 60% more lifetime non-housing consumption to match the 1940 environment. Earnings and house prices are the two channels that carry it, earnings the larger of the two, and interest rates are close to zero.[21]

That aggregate covers a wide range, and the cohort can be sorted in two ways. Figure 11 opens with the whole-cohort number on the left and then splits the loss by what a household holds at age 25. Figure 12 splits it by what the household earns.

[21]The relative size of the earnings and price contributions depends on the income path assigned to the 1980 cohort over the ages it has not yet reached. See the corresponding note in Section V.A.3.

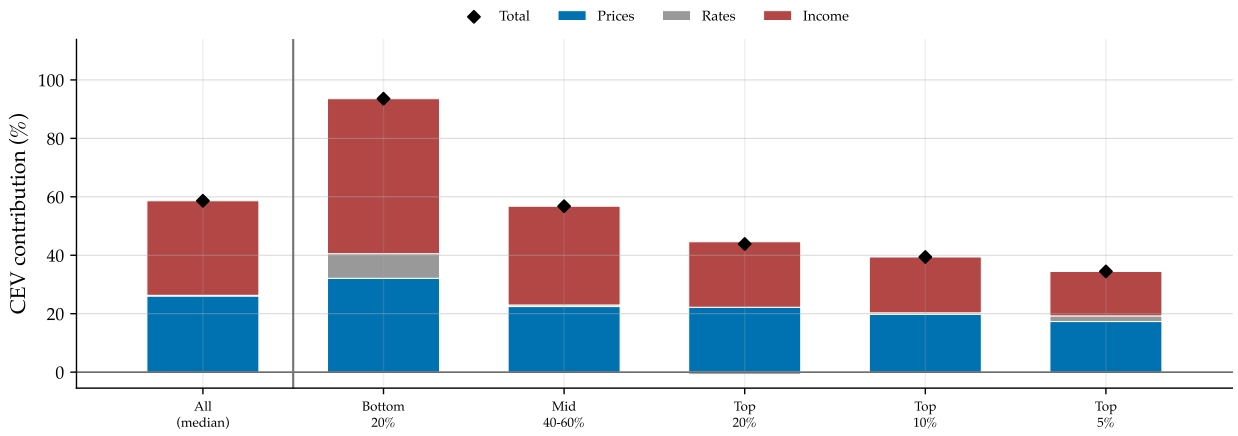


Figure 11. Ex-ante Welfare Loss: Whole Cohort and by Initial Wealth

Notes: CEV (%) of the 1980 → 1940 environment change, three market factors $\{P, R, Y\}$. The left bar is the whole-cohort median loss; the remaining bars are five groups of age-25 net worth, and the three upper groups are nested. Shapley is applied to each group's median CEV, so the contributions sum to the total exactly. Contributions stack and the diamond marks the total.

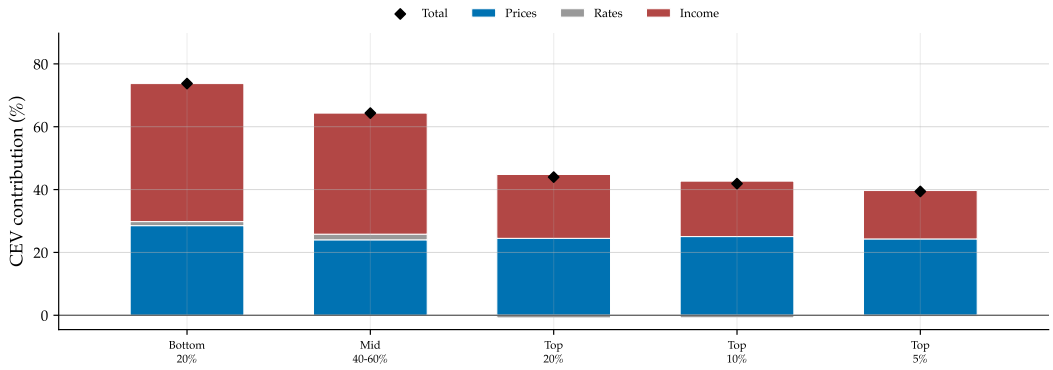


Figure 12. Ex-ante Welfare Loss by Initial Income

Notes: CEV (%) of the 1980 → 1940 environment change, three market factors $\{P, R, Y\}$. As Figure 11, over five groups of age-25 pre-tax income.

Set against Section V.A.3, the composition of the loss is informative. House prices dominate the ownership gap at every age, but in the welfare calculation earnings carry at least as much weight, and at the bottom of the wealth distribution considerably more.

3. *Who Bears It More*

The wealth cut is the steeper of the two. Sorted on wealth, the loss falls from 94% in the bottom fifth to 35% in the top 5%. Sorted on income, the same two quantiles run from 74% to 39%. What separates the two cuts is in the channels. Along the wealth distribution both give way: the price contribution falls from 32 to 17 and the earnings contribution from 53

to 15, so a household at the top carries less of each. Along the income distribution only one of them moves. The earnings contribution falls from 44 to 15, much as it does with wealth. The price contribution is 28 in the bottom fifth and 24 in the top 5%, and it moves little in between.

A higher income lowers the loss that comes through earnings, and moves the loss that comes through house prices only slightly. More wealth lowers both. What the wealth cut is picking up shows in the owners. Sorting owners by wealth is close to sorting them by how much of the house was financed with debt. A house is a hedge against the cost of housing. A mortgage is leverage laid on top of that hedge, and where the loan-to-value is high the leverage dominates. The environment is unforgiving to an owner who borrowed heavily at age 25.

The loss therefore changes character across the income distribution. In the bottom fifth the earnings channel carries most of it. In the top 5% house prices do.

Table 5 carries the cell-level detail behind both figures.

Table 5. Welfare-loss Decomposition by Representative Group

	Shapley CEV (%)			
Group	Prices	Rates	Income	Total
Panel A. Renters — by income & quality				
tier1, Low	+25.5	+1.0	+44.0	+70.5
tier1, Mid	+23.5	−0.1	+33.6	+57.0
tier1, High	+25.3	−1.1	+22.3	+46.5
tier2, Low	+67.0	+4.5	+75.2	+146.6
tier2, Mid	+29.0	+2.9	+33.7	+65.6
tier2, High	+20.7	+0.5	+20.7	+41.8
tier3, Low	+78.2	+10.5	+124.0	+212.7
tier3, Mid	+54.9	+4.6	+55.9	+115.4
tier3, High	+27.6	+3.0	+25.7	+56.3
Panel B. Owners — by income, quality & leverage				
tier1, Low, Leveraged	+88.5	−5.3	+57.6	+140.8
tier1, Low, Outright†	+9.4	+1.1	+19.9	+30.4
tier1, Mid, Leveraged	+28.1	−4.7	+32.4	+55.8
tier1, Mid, Outright	+8.9	+1.0	+18.7	+28.5
tier2, Mid, Leveraged	+70.9	−9.0	+29.4	+91.3
tier2, Mid, Outright	+14.7	+1.9	+17.0	+33.6
tier2, High, Leveraged	+36.9	−4.8	+17.8	+49.8
tier2, High, Outright	+12.3	+3.3	+13.4	+29.0
tier3, High, Leveraged	+39.1	−6.0	+21.7	+54.7
tier3, High, Outright	+16.4	+4.8	+10.8	+32.0
Panel C. By initial wealth				
Bottom 20%	+32.1	+8.4	+53.0	+93.6
Mid 40–60%	+22.4	+0.5	+33.8	+56.8
Top 20%	+22.2	−0.8	+22.4	+43.9
Top 10%	+19.8	+0.5	+19.1	+39.4
Top 5%	+17.3	+1.9	+15.2	+34.5

Notes: Shapley CEV (%) of the 1980 → 1940 environment change, three market factors $\{P, R, Y\}$, applied to each group's median CEV so the contributions sum to the total exactly. Panel A splits renters by band of age-25 pre-tax earnings (Low $\leq$ \$35k, Mid \$35–60k, High $>$ \$60k) and by the quality tier they start in. Panel B splits owners by the same income bands, tier, and leverage (leveraged is initial LTV ≥ 0.6, outright is LTV $= 0$), keeping only groups that contain both. Panel C is the five wealth groups of Figure 11. Groups with fewer than 30 households are dropped.

Low earnings and high leverage are what decide who loses. Both are markers of starting life with fewer resources, and both point the loss toward the households least able to bear

it. Read in reverse, the incidence of the environment that younger cohorts actually face is sharply regressive: the cost falls hardest on those who began with least, and it does so on the ownership margin, the consumption margin, and in the welfare aggregate alike. Our age- and tenure-incidence echoes Fagereng et al. (2025): rising asset valuations transfer resources from young net buyers to old net sellers, with housing the largest channel.

VI. Robustness

The headline results hold the environment at its default paths beyond the data window: the mid income projection for the 1960 and 1980 cohorts, the baseline house-price path, and the time-varying interest rates each cohort faced. Two of these are assumptions the data cannot pin down, so we vary them one at a time around the baseline. We re-solve the model under a conservative and an optimistic income projection, and we trace the 1980 cohort's late-life outcomes under alternative house-price paths. Interest rates enter only through a footnote, because the decomposition assigns them a near-zero share of the loss.[22]

VI.A. Income Projection

The 1980 cohort, and the 1960 cohort at its oldest ages, are assigned an income path over years the data do not yet cover. Our default is a mid projection. We bracket it with a conservative path (stagnation continues) and an optimistic one (younger cohorts catch up), holding house prices and interest rates at baseline. Only the 1960 and 1980 income targets move. The 1940 cohort and the calibration are untouched, so the comparison isolates the projection.

Table 6. Homeownership Gap and Welfare Loss under Alternative Income Projections

Income scenario	HO gap, 25–34 (pp)	Welfare loss (%)	of which (P+R+Y)		
			Prices	Rates	Income
Conservative (LOW)	25.0	64.2	26.2	0.7	37.3
Default (MID)	**25.8**	**58.6**	**26.0**	**0.3**	**32.3**
Optimistic (HIGH)	27.3	52.6	25.5	0.1	27.1

Notes: Young-cohort homeownership gap (ages 25–34, 1940 minus 1980, in percentage points) and aggregate ex-ante welfare loss (median CEV, three market factors $\{P, R, Y\}$, with the price, rate, and income split, which is additive: $P + R + Y$ equals the total). House prices and rates held at baseline; only the 1960/1980 income projection varies.

[22] Re-solving the model with constant interest rates (in real-term) throughout moves young homeownership by only 1–5 percentage points, more so under the higher sample-average borrowing cost, and leaves later-life outcomes essentially unchanged since most households have repaid their mortgages by retirement.

Under the default projection the young homeownership gap is 25.8 points and the aggregate welfare loss is 58.6% of lifetime non-housing consumption. The two respond differently to the income assumption. The welfare loss moves with it, from 52.6% under the optimistic path to 64.2% under the conservative one, and the movement runs almost entirely through the earnings channel; the price contribution is essentially unchanged across scenarios (about 26% in each), so prices remain a first-order force whatever earnings do. The homeownership gap, by contrast, barely moves (25–27 points), because it is set mainly by prices and by early-career earnings, which are common to all three paths; if anything it is marginally wider under the optimistic path, where households expecting higher future income save and buy a little less when young. Two things travel across the whole band: prices carry a large, stable share, and even under the optimistic projection the welfare loss exceeds half of lifetime non-housing consumption. A more conservative outlook only widens it, so the headline is best read as a lower bound.[23]

VI.B. House Price Scenarios

The 1980 cohort's retirement outcomes depend on prices not yet observed. We consider three projections from 2025 onward, holding all other parameters at their calibrated values:

- *Baseline*: 1% real annual growth, the post-war long-run average.
- *Boom-bust*: the 2002–2020 pattern of rapid appreciation and sharp correction, repeated.
- *Gradual rise*: the mid-1990s pattern of steady gains, repeated.

Because the paths diverge only from age 45, entry is barely affected: homeownership through the mid-40s is nearly the same across scenarios, within one percentage point at age 35 and two at age 45. From age 45 the differences are modest. Under the gradual-rise path homeownership runs slightly higher through mid-life (about 57% at ages 45–54 against 55% in the baseline), then falls furthest after age 65 (about 47% against 52%): higher prices raise the return to selling, and retirees sell into the elevated market, so the path that lifts mid-life ownership depresses it in old age while leaving retirees with more housing wealth (median net worth near age 65 about $327k against $272k in the baseline). The boom-bust path is a muted version of the same pattern.

In the model, late-life consumption is funded only by running down a single saving asset or by selling the house, so once liquid saving is gone the home is the only wealth left

[23]Interest rates enter only through their default time-varying path. The welfare decomposition assigns rates a near-zero share of the loss (Section V.C), so replacing the time-varying path with a constant one leaves the results essentially unchanged; we do not treat rates as a separate scenario.

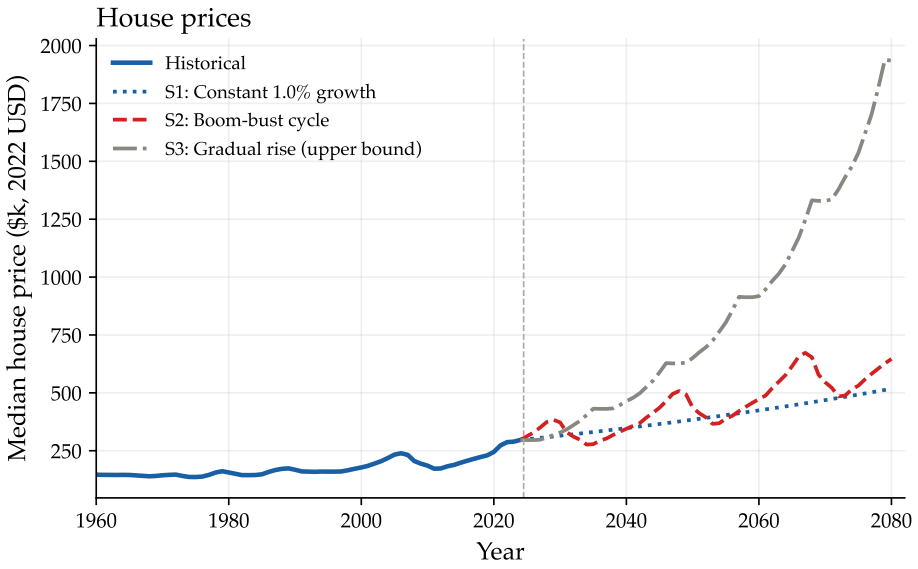


Figure 13. Medium House Prices: Historical and Projection Scenarios

Notes: Historical series from the Median Sales Price of Houses Sold (FRED series MSPUS), deflated by CPI to 2022 dollars and scaled to medium-quality (Q2) housing. Scenario 1: constant 1.0% annual real growth. Scenario 2: repeat of the 2002–2020 boom-bust cycle. Scenario 3: repeat of the mid-1990s trajectory.

to draw on. Households in fact have more margins than this. They hold financial portfolios that the model compresses into that one asset, and they spend down retirement-account balances (401(k) and IRA wealth) that the model carries as an annuity-like pension flow rather than a stock one can cash out. Both let retirees finance old age without touching the house. Equity-release products do the same against the home directly, and although take-up is still a small share of eligible households it has been rising. Because the model omits these channels, it routes more of late-life decumulation through house sales than reality would, so the projected post-65 ownership decline is best read as an upper bound, most so in the high-appreciation scenarios where the incentive to sell into an elevated market is strongest.

Table 7. House Price Scenarios (1980 cohort): Homeownership and Wealth by Age

Panel A. Homeownership rate (%) Scenario	Age 35	Age 45	Age 55	Age 65
S1: Constant 1.0% growth (baseline)	47.6	55.4	57.1	52.2
S2: Boom-bust cycle	48.0	56.2	58.0	51.7
S3: Gradual rise	48.3	57.2	58.6	46.8
Panel B. Median net worth ($k) Scenario	Age 35	Age 45	Age 55	Age 65
S1: Constant 1.0% growth (baseline)	19	155	276	272
S2: Boom-bust cycle	20	170	246	290
S3: Gradual rise	20	167	338	327

Notes: 1980 cohort. Historical house prices up to 2024 (age 44); scenarios differ only in the projected path after 2024. S1 (constant 1.0% real growth) is the baseline used in the main results.

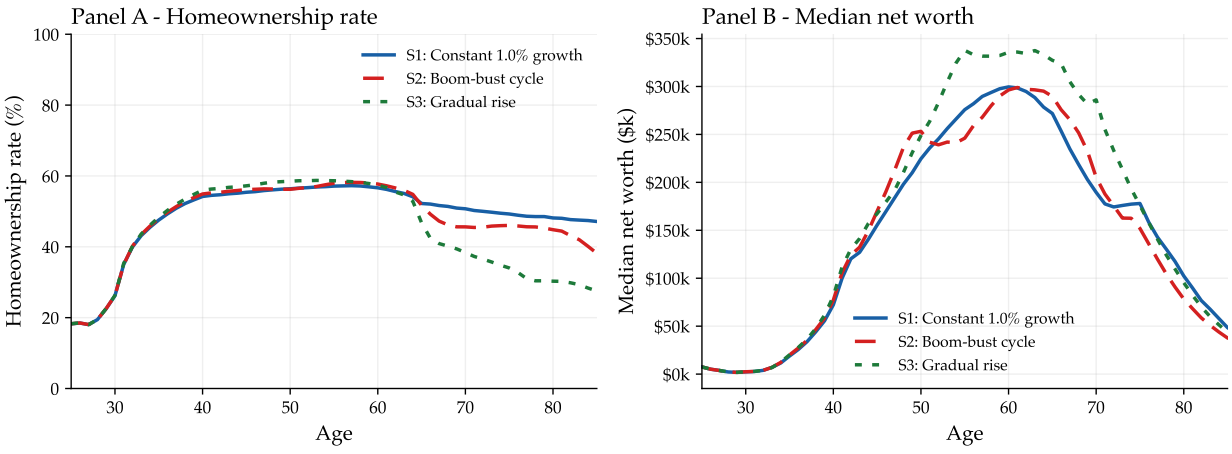


Figure 14. Homeownership and Median Wealth under Alternative House Price Paths

Notes: 1980 cohort. Panel A: homeownership rate; Panel B: median net worth. Historical prices (ages 25–44) are identical across scenarios; the projections diverge from age 45 onward.

VI.C. Household Size

The baseline model does not account for household size. Utility in equation (6) does not depend on how many people live in the household, so a given level of non-housing consumption yields the same utility in every cohort. Later cohorts, however, live in smaller households when young. As a check, we let the 1980 cohort value non-housing consumption per unit of household size relative to the 1940 cohort:

$$u_t(c, h', q') = \frac{(c/\eta_t)^{1-\gamma} - 1}{1-\gamma} + \phi_q \frac{g(q')^{1-\psi} - 1}{1-\psi} + \nu \cdot \mathbb{1}\{h' = 1\}, \tag{21}$$

where η_t is the ratio of the 1980 cohort's average household size to that of the 1940 cohort at the same age, measured with the original OECD equivalence scale. We observe the 1980 cohort up to age 43, where the ratio rises from about 0.76 at age 25 to about 0.93. We fit a logistic curve in age to these observed ratios and extend it to later ages, where it approaches one by the mid-fifties, as the observed ratio of the 1960 cohort does. The baseline and the 1940 cohort have $\eta_t = 1$. Housing services, the preference parameters and the calibration are left unchanged.

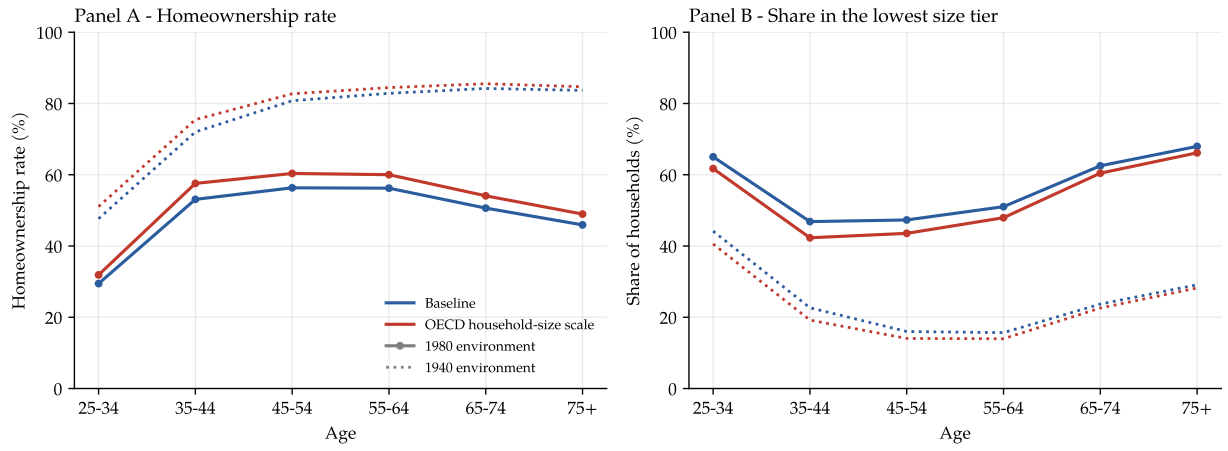


Figure 15. Homeownership and Housing Size with a Household-Size Adjustment

Notes: 1980 cohort, averages by age band. Solid lines: the 1980 environment. Dotted lines: the same cohort given the 1940 house prices, interest rates and earnings. Baseline: utility does not depend on household size. OECD scale: utility as in equation (21). Panel B: share of households, owners and renters, in the lowest size tier.

With smaller households, the same consumption goes further, so the 1980 cohort owns somewhat more, by 2.4 percentage points at ages 25 to 34 and by about 4 points at later ages, and fewer of its households stay in the lowest tier (Figure 15). The young homeownership gap narrows from 25.8 to about 23 points. The effect of the environment is unchanged. Giving the 1980 cohort the 1940 house prices raises its ownership at ages 25 to 34 by 13.5

percentage points, against 13.8 in the baseline, and the distance between the solid and dotted lines is almost the same under both specifications at every age. Our conclusions on ownership, housing size and the role of house prices are therefore robust to differences in household size across cohorts.

VII. Conclusion

We asked how much of the changing housing experience across the 1940, 1960, and 1980 cohorts the environment alone can carry. Holding preferences fixed at the 1940 calibration and moving only the exogenous inputs (prices, rates, earnings, and survival), the model reproduces most of it: later and lower entry into ownership, smaller homes, non-housing consumption that stalls for the young, and welfare losses that land on the later-born (Section V.C). We read this as sufficiency. Tastes and demographics need not have stood still, but one does not have to invoke them to get the facts. The environment already delivers the outcome, so any story that leans on changing preferences has to earn that extra step.

The gap this opens is generational. Each later cohort spends more of its life inside a tighter version of the same environment, and the disadvantage does not wash out with age: entering late and constrained leaves a mark that compounds through the whole tenure and wealth history. Two forces drive it, working in opposite directions on the same household. House prices rise against a supply that zoning, regulation, and developer caution hold back, while entry wages leave the young furthest from the homes they are trying to reach. Earlier cohorts bought before the climb and gained from it. Later ones arrive with less to bargain with in either market and do not catch up.

What this does to the cohorts still to come is the open question. The squeeze falls most heavily in young adulthood, the years when people leave home and start families. Causation is usually traced from demographics to housing, and some of it does. But the mechanism can also run in the other direction: housing conditions may affect household formation, which these accounts treat as fixed. We measure the squeeze, not this feedback, so we leave it as an open thread. This suggests that today's cohort gap may have consequences beyond housing.

Two limits set the agenda from here. The exercise is partial equilibrium, silent on how supply or prices would answer a policy. In general equilibrium higher rates would also depress house prices while rents adjust less (Sommer et al., 2013), a feedback we shut down. And the rent-assistance rule is modelled as universal while the real program is rationed, so the renter losses are, if anything, too small. Across cohorts, the housing and

economic environment faced in early adulthood has become more important for later-life outcomes. Whether that pressure also delays marriage and childbearing, and so shapes the cohort that follows, is the question we would put next.

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Appendix

A. Sample and Descriptive Statistics

Table 8 reports weighted PSID descriptive statistics at ages 25–34 for the three cohorts. The demographic backdrop shifts markedly across them: the share of female-headed households rises from 17% to 31%, average family size falls from 3.2 to 2.3, and the fraction

with any children drops from 69% to 41%. College completion is flat between the 1940 and 1960 cohorts (around 24%) before jumping to 39% for the 1980 cohort.

Real median family income falls from \$72,000 to \$55,000 across the two cohorts, a decline concentrated in the lower half of the distribution: the mean falls far less, from \$78,700 to \$70,000. Employment weakens at the margin, with the unemployment rate rising from 3% to 9% and the not-in-labour-force share edging up.

The housing picture changes most sharply. Homeownership at ages 25–34 falls from 53% to 31%, even as the real median home value conditional on owning rises by roughly 40% (\$148,000 to \$207,000). Younger cohorts thus face both a lower probability of owning and a higher price conditional on buying, and the typical home is smaller, with the average room count falling from 5.3 to 4.1.

B. Survival Probabilities

Each cohort faces its own conditional survival schedule $\{\pi_t\}_{t=1}^{T-1}$, where π_t is the probability of surviving from period t to $t+1$ (period $t=1$ is age 25). We take these from the SSA cohort life tables and form a population-weighted unisex average, so that the schedule reflects the sex composition of each birth cohort rather than a single-sex table. Agents enter at age 25, retire at 65, and face a cohort-specific terminal age that rises with longevity across cohorts: 90 for the 1940 cohort, 92 for 1960, and 94 for 1980 (two-year steps). The horizon T is therefore 66, 68, and 70 model periods, respectively.

The terminal age governs late-life decumulation and the timing of bequests rather than the working-age homeownership moments, which are all reached well before age 74. In the Shapley market decomposition (Appendix H) survival is held at its 1980 schedule for all configurations, so that the decomposition isolates the three market channels; the separate role of longevity is discussed in the main text.

C. Earnings Process

Pre-tax labour earnings follow

$$\ln y_{it}^{\text{pre}} = \alpha_0 + \alpha_1 \,\text{age}_t + \alpha_2 \,\text{age}_t^2 + s_{it}, \qquad s_{it} = \rho\, s_{i,t-1} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma_{\varepsilon,t}^2), \tag{22}$$

with persistence $\rho = 0.92$ common across cohorts, within the 0.90–0.98 range used for comparable annual processes (Flöden, 2001; Chang, 2024; Krueger and Ludwig, 2007). The deterministic coefficients $(\alpha_0, \alpha_1, \alpha_2)$ are pinned down, cohort by cohort, by fitting a quadratic in age to the log of median real labour earnings at ages 30, 50, and 60 from the

Table 8. Cohort Descriptive Statistics, Ages 25–34

	1940s	**1960s**	**1980s**
Panel A: Sample			
N (unweighted)	12,828	19,492	11,856
N (pop. $\sum w_i$)	197,794	264,850	182,345
Panel B: Demographics			
Age of head (mean)	29.8	29.3	29.8
Couple (married or cohabiting)	70.1%	53.2%	44.5%
Female head	16.6%	28.3%	30.5%
Panel C: Education			
Years of education (mean)	13.3	13.2	14.0
College or above ($\geq$16 yrs)	24.1%	24.4%	39.0%
Panel D: Family structure			
Family size (mean)	3.2	2.5	2.3
Number of children (mean)	1.43	0.91	0.80
Has any children	69.2%	47.7%	40.8%
Panel E: Income			
Family income (mean)	\$78,697	\$70,113	\$69,980
Family income (median)	\$72,530	\$58,575	\$54,956
Panel F: Employment			
Employed	91.1%	88.1%	84.3%
Unemployed	3.2%	5.6%	8.5%
Not in labor force (NILF)	5.6%	6.3%	7.3%
Panel G: Housing			
Homeownership rate	53.4%	37.7%	31.4%
Home value mean (if owner)	\$166,808	\$180,998	\$251,996
Home value median (if owner)	\$148,428	\$149,541	\$207,277
Number of rooms – all (mean)	5.3	4.9	4.1

Notes: Weighted using PSID longitudinal family weights (population-representative). N (pop.) $= \sum w_i$. Observation windows. 1940s: 1968-1983; 1960s: 1985-2003; 1980s: 2005-2023. Partial coverage means the cohort's age window extends beyond the panel's 1968–2023 span. *Couple* = head living with a legal spouse or a cohabiting partner. *Employed* includes the self-employed and those on temporary leave; *NILF* = retired, disabled, keeping house, student, or other. Monetary values in 2022 USD (CPI). *Source:* PSID Main Family Data File, 1968–2023.

PSID (Panel A of Table 9); labour earnings are approximated as a fixed fraction, 0.92, of PSID family income, a measurement adjustment unrelated to the persistence parameter above. Because the median of a log-normal equals the exponential of its conditional mean, these three points exactly identify the three coefficients.

The innovation standard deviation at labour-market entry, $\sigma_{\varepsilon,1}$, is backed out from the cross-sectional standard deviation of log earnings at age 30, using the stationary relation $\mathrm{Var}(s) = \sigma_{\varepsilon,1}^2/(1-\rho^2)$. The resulting $\sigma_{\varepsilon,1}$ rises across cohorts ($0.157 \to 0.176 \to 0.196$), reflecting the well-documented increase in U.S. earnings inequality. Sommer et al. (2013) read the PSID evidence as putting the innovation standard deviation between 0.12 and 0.25 and the persistence between 0.88 and 0.96. Our values for both fall inside those bands. Within each cohort the innovation variance also widens over the working life,

$$\sigma_{\varepsilon,t} \;=\; \sigma_{\varepsilon,1}\sqrt{1+\zeta\,\frac{t-1}{T_{\mathrm{ret}}-2}}\,, \qquad \zeta = 1.8, \qquad t < T_{\mathrm{ret}}, \tag{23}$$

so the innovation standard deviation rises from $\sigma_{\varepsilon,1}$ at labour-market entry to $\sqrt{1+\zeta}\,\sigma_{\varepsilon,1}$ in the last working year; for the 1940 cohort, from 0.157 to 0.262. The cross-sectional spread of earnings therefore rises with age, as it does when the PSID cohorts are followed over their lifecycle. For the numerical solution the process is discretised into $N_s = 25$ points by the method of Tauchen (1986) on a grid four standard deviations wide. The grid is built once, at the widest innovation variance the working life reaches, and only the transition matrices vary with age. The reported figures use the baseline (MID) income targets; the LOW and HIGH variants that move the 1960 and 1980 targets are studied in Section VI. Retirement income is a deterministic function of last working-period earnings (Appendix E).

D. The Quality Ladder

Utility is concave in quality, so a ladder of tiers runs into a ceiling. At the estimated curvature, if the utility index were the observed value ladder itself, $g(q) = q$, the top two tiers would be almost indistinguishable: tier 2 already delivers 98% of the housing utility available at any quality, and tier 3 delivers 99.9%. The model would place almost no one in the largest home, whatever the data show.

We therefore set g so that the tiers stay apart, imposing that the utility gain from moving up a tier is proportional to the log of the ratio of the two tiers' values:

$$u(q_i) - u(q_j) \;=\; A\,\ln\!\left(\frac{q_i}{q_j}\right). \tag{24}$$

Table 9. Earnings Process — Target Moments and Estimated Parameters

Panel A: Target moments (PSID labour earnings, $k 2022 real)				
	Median y_{30}	Median y_{50}	Median y_{60}	SD($\log y_{30}$)
1940	66	86	70	0.40
1960	60	72	65 [60–70]	0.45
1980	53	75 [64–90]	60 [50–72]	0.50
Panel B: Estimated parameters				
	α_0	α_1	α_2	$\sigma_{\varepsilon,1}$
1940	2.1016	0.1034	−0.001127	0.1568
1960	2.8537	0.0607	−0.000645	0.1764
1980	1.4658	0.1232	−0.001322	0.1960

Notes: $\rho = 0.92$ for all cohorts. Target medians are in thousands of 2022 dollars (labour earnings $\approx 0.92\times$ PSID family income), under the baseline (MID) income scenario. Bracketed ranges are the LOW–HIGH band for the three targets not yet observed (1960 at age 60, in 2020; 1980 at ages 50 and 60, in 2030 and 2040); unbracketed targets are observed and common to all scenarios, and Section VI.A reports results across the band. The coefficients $\alpha_0, \alpha_1, \alpha_2$ solve $\log(\text{median}_a) = \alpha_0 + \alpha_1 a + \alpha_2 a^2$ at $a \in \{30, 50, 60\}$, and $\sigma_{\varepsilon,1} = \text{SD}(\log y_{30}) \times \sqrt{1-\rho^2}$; this is the value at labour-market entry, from which $\sigma_{\varepsilon,t}$ widens over the working life.

This is not the only rule that would do the job. What any alternative would have to share is the requirement that no tier be left dominated. A quality level no household would ever choose has an observed share the model cannot be asked to match. What follows is one way of meeting that requirement.

Since $q = (1,\ 2,\ 3.1)$, the ladder rises with diminishing gains: the step from the medium to the large home is worth $\ln(3.1/2)/\ln 2 = 0.63$ of the step into the medium one, a ratio fixed by the observed values alone. Given the first two points of the index, (24) and the utility function in (6) determine the third: setting $g(q_1) = 1.3$ and $g(q_2) = 1.5$ gives $g(q_3) = 2.08$. The first two points are not unique, and nothing in the results turns on them. A different admissible pair reproduces the same ladder, and hence the same policy functions, simulated moments and welfare figures, once the remaining parameters are re-solved. Table 10 reports one such pair alongside the baseline.

E. Retirement Income

Retirement income for an agent with gross final working-period earnings y^{pre} and post-tax earnings y^{post} is the sum of Social Security benefits and private retirement income.

Table 10. The quality ladder and its normalisation

	tier 1	tier 2	tier 3
q_j (PSID median house value, tier 1 = 1)	1.00	2.00	3.10
$g(q_j)$, baseline	1.30	1.50	2.08
$g(q_j)$, alternative	1.15	1.30	1.45
Largest difference across the thirteen simulated moments: 4×10^{-15}			

Notes: q enters the budget; $g(q)$ enters utility. The two configurations differ only in the first two points of the index, with the remaining parameters re-solved to hold the utility ladder fixed.

Social Security We apply the statutory PIA formula to an approximation of Average Indexed Monthly Earnings (AIME),

$$\text{AIME} = \frac{\min(y^{\text{pre}}, \bar{y}^{SS}) \times 1000}{12}, \tag{25}$$

where $\bar{y}^{SS}$ is the cohort-specific taxable maximum (Table 11). The Primary Insurance Amount is

$$\text{PIA} = 0.90\ \min(\text{AIME}, B_1) + 0.32\ \min\big(\max(\text{AIME} - B_1, 0), B_2 - B_1\big) + 0.15\ \max(\text{AIME} - B_2, 0), \tag{26}$$

with bend points B_1, B_2 for the year the worker turns 62. Annual Social Security income is $R^{SS} = \text{PIA} \times 12/1000$ in model units of \$k.

Private retirement income Private retirement income R^{priv} applies a replacement rate to post-tax final earnings that rises with income, anchored to combined DC/IRA replacement rates (Employee Benefit Research Institute, 2025).

Total Pre-tax retirement income is $R^{\text{pre}} = R^{SS} + R^{\text{priv}}$, subject to a floor of $\underline{b}$ (the safety-net transfer) and a ceiling of $100\,\underline{b}$. Post-tax retirement income is $R^{\text{post}} = \lambda\,(R^{\text{pre}})^{1-\tau_r}$ with $\tau_r = 0.10$.

E.1. Social Security Parameters by Cohort

Table 11 summarises the cohort-specific Social Security parameters. For the 1940 and 1960 cohorts, bend points are the statutory values for the year the worker turns 62, deflated to 2022 dollars with the CPI. For the 1980 cohort they are projected from the 2022 base at 0.58% annual real wage growth, the average real growth of the taxable maximum over

2010–2023.

Table 11. Social Security Parameters by Cohort (2022 Real Dollars)

Cohort	Year turns 62	B_1 (\$/month)	B_2 (\$/month)	Taxable Max (\$k)	Method
1940	2002	963	5,803	135	CPI deflation
1960	2022	1,024	6,172	147	Base year
1980	2042	1,150	6,929	168	0.58%/yr projection

Notes: Sources: Bend points from SSA ; taxable maximum from SSA Nominal values deflated to 2022 dollars using the BLS CPI annual average (CPI_{2022} = 292.655). The projection rate is the average real growth of the taxable maximum over 2010–2023.

F. Effective Tax Rate Schedule

Table 12 reports the effective average tax rates implied by the HSV tax function for selected gross income levels. During working years the function $T(y) = y - \lambda\, y^{1-\tau_w}$ with $\lambda = 1.50$ and $\tau_w = 0.18$ generates rates that rise from near zero at \$10,000 to 57% at \$1,000,000, broadly in line with the federal average tax rates (individual income plus payroll) the CBO reports rising with income (Congressional Budget Office, 2024). During retirement the lower progressivity parameter $\tau_r = 0.10$ reflects the favourable tax treatment of Social Security and pension income (Chen and Munnell, 2020).

Table 12. Implied Effective Tax Rates by Income Level

	Working Period		Retirement	
Gross Income (\$k, 2022)	Post-tax (\$k)	Eff. Rate (%)	Post-tax (\$k)	Eff. Rate (%)
5	6	−12	7	−30
10	10	1	12	−16
30	24	19	31	−3
50	37	26	47	6
100	65	35	85	15
200	116	42	157	22
500	245	51	348	30
1,000	433	57	637	36

Notes: Tax function is $T(y) = y - \lambda\, y^{1-\tau}$ with $\lambda = 1.50$; working $\tau_w = 0.18$, retirement $\tau_r = 0.10$. For comparison, CBO average federal tax rates (individual income plus payroll) also rise with income over this range (Congressional Budget Office, 2024).

G. Solution Method

The model is solved by backward induction using the discrete-continuous endogenous grid method (DC-EGM) of Iskhakov et al. (2017). In each period an agent chooses non-housing consumption on a continuous asset grid together with two discrete choices: tenure (rent or own, $N_h = 2$) and, for owners, housing quality (small, medium, or large, $N_q = 3$). Conditional on the discrete choices, the Euler equation is inverted on an exogenous grid of end-of-period assets to recover the consumption policy, which avoids root-finding within the continuous problem.

Because the discrete choices make the conditional value functions non-concave, the endogenous consumption correspondence can be non-monotone. We resolve this with the upper-envelope step of Iskhakov et al. (2017): candidate segments are compared pointwise and dominated segments discarded, with the crossing points located by interpolation. The discrete choice is then taken as the arg-max over the resulting conditional value functions.

The asset grid uses $N_a = 300$ points spanning \$-1,200 (the borrowing limit) to \$5,000 thousand, denser near the constraint; the income process is discretised into $N_s = 25$ Tauchen nodes (Appendix C). The same grids are used for every cohort and every counterfactual configuration, so that differences across solutions reflect the changed inputs rather than numerical resolution.

H. Shapley Decomposition of the Cohort Gap

Throughout the paper we ask one counterfactual question: of the gap between the 1980 and 1940 cohorts, how much is due to house prices, how much to interest rates, and how much to the earnings process? We answer it by moving these three channels from their 1980 values to their 1940 values and re-solving the model, not one at a time, but in every combination. Switching one channel at a time would not do, because the model is nonlinear and the channels interact. Cheaper houses raise ownership by more when earnings are also stronger; a lower borrowing rate matters only once a household has decided to buy. The contribution one assigns to a channel therefore depends on which others have already been switched, and the single-channel swaps do not add up to the total gap.

The Shapley value is the standard remedy. It gives each channel the average of its marginal contribution taken over every order in which the channels could be switched on. With three channels there are $2^3 = 8$ combinations—the pure 1980 baseline, the full 1940 environment, and the six in between—and each is a complete solve-and-simulate of the model. Write $\mathcal{N} = \{P, R, Y\}$ for the three channels, $S \subseteq \mathcal{N}$ for a combination switched to 1940 (the rest left at 1980), and $O(S)$ for the outcome under that combination:

the homeownership rate in Section V.A.3, the welfare loss (CEV) in Section V.C.2. The contribution of channel f is

$$\phi_f = \sum_{S \subseteq \mathcal{N} \setminus \{f\}} \frac{|S|!\,\big(3 - |S| - 1\big)!}{3!} \big[\, O(S \cup \{f\}) - O(S) \,\big], \tag{27}$$

where the weight is the share of orderings in which exactly the channels in S precede f.

The reason for proceeding this way is that the pieces add up. For any outcome O, however nonlinear the model that produces it,

$$\phi_P + \phi_R + \phi_Y = O(\mathcal{N}) - O(\varnothing), \tag{28}$$

the full 1940-minus-1980 gap. The interaction between channels is not thrown away; it is shared evenly among the channels that create it, so the three contributions sum to the total by construction and can be read as an exact attribution. The same simulation seed is used for every combination, so the differences $O(S \cup \{f\}) - O(S)$ net out common Monte-Carlo noise.

For the welfare outcome, $O(S)$ is the cross-sectional median per-agent CEV under configuration S, computed once per configuration exactly as for the homeownership rate, so that efficiency continues to hold exactly. We report the median because the per-agent CEV distribution is right-skewed, with a small mass of high-quality renters whose near-zero baseline utility inflates the mean; the median is robust to this pole, while the utility-aggregate alternative gives similar channel shares.

We hold the rest of the environment (survival, the retirement horizon, Social Security, and the age-25 initial distribution) at its 1980 values throughout, so the decomposition speaks to the three market channels a household actually faces. The same procedure extends to those features; the fuller multi-factor version and its implementation are documented alongside the replication code. The Shapley value is due to Shapley (1953), and Shorrocks (2013) develops its use for decomposing distributional statistics.

I. Ex-ante Welfare: Derivation of the Consumption-Equivalent Variation

The measure of Section V.C.1 is the ex-ante consumption-equivalent variation, standard in the welfare literature (Flöden, 2001; Conesa et al., 2009; Sommer and Sullivan, 2018): the proportional change in non-housing consumption that equates a household's ex-ante expected lifetime value across the two environments, $W_i^A\big|_{c \to (1+\lambda_i)c} = W_i^B$, with A the baseline (1980) environment and B the alternative (1940). Our only addition is the closed

form for a lifecycle model with additively-separable housing utility and survival weighting. We record the derivation because one step needs care: the CRRA normalisation constant cancels when consumption is scaled and so does not enter the denominator $A_{0,i}$.

Utility and its consumption block Per-period utility is the normalised CRRA form used throughout,

$$u(c,h,q) = \frac{c^{1-\gamma}-1}{1-\gamma} + \phi_q \frac{q^{1-\psi}-1}{1-\psi} + \nu\, \mathbb{1}\{h=1\}, \qquad \gamma > 1. \tag{29}$$

Split the consumption utility into the part that carries the consumption level and a constant,

$$\frac{c^{1-\gamma}-1}{1-\gamma} = \underbrace{\frac{c^{1-\gamma}}{1-\gamma}}_{a_c(c)} - \frac{1}{1-\gamma}. \tag{30}$$

Only $a_c(c)$ moves with c; the constant $-1/(1-\gamma)$, the quality and ownership terms, the switching cost, and the warm-glow bequest do not. With $t = 1, \ldots, T$ indexing model age ($t = 1$ is age 25), let S_t be the cohort-common probability of surviving to t and W_i^e the ex-ante expected lifetime value under environment $e \in \{A, B\}$. Define

$$A_{0,i} \equiv \mathbb{E}_i^A\left[\sum_{t=1}^{T} \beta^{t-1} S_t\, a_c\big(c_{it}^A\big)\right] = \mathbb{E}_i^A\left[\sum_{t=1}^{T} \beta^{t-1} S_t \frac{(c_{it}^A)^{1-\gamma}}{1-\gamma}\right], \qquad H_i \equiv W_i^A - A_{0,i}, \tag{31}$$

so that $W_i^A = A_{0,i} + H_i$. By (30), H_i collects every consumption-level-independent term, including the $-1/(1-\gamma)$ constants and the bequest; $A_{0,i}$ is built from the power term a_c alone.

Step 1: scaling consumption Raise the baseline consumption stream by a factor $x > 0$, holding all other choices fixed. The change in per-period consumption utility, evaluated with the full normalised utility, is

$$\frac{(xc)^{1-\gamma}-1}{1-\gamma} - \frac{c^{1-\gamma}-1}{1-\gamma} = \frac{x^{1-\gamma}c^{1-\gamma} - c^{1-\gamma}}{1-\gamma} = \big(x^{1-\gamma}-1\big)\frac{c^{1-\gamma}}{1-\gamma}, \tag{32}$$

so the constant -1 cancels and only the power term survives. Summing against $\beta^{t-1}S_t \geq 0$ and taking expectations, which is linear,

$$W_i^A\Big|_{c\to xc} = x^{1-\gamma} A_{0,i} + H_i. \tag{33}$$

Step 2: solving for λ_i The compensating variation is defined by $W_i^A\big|_{c\to(1+\lambda_i)c} = W_i^B$. Setting $x = 1+\lambda_i$ in (33) and substituting $H_i = W_i^A - A_{0,i}$,

$$(1+\lambda_i)^{1-\gamma} A_{0,i} + W_i^A - A_{0,i} = W_i^B, \tag{34}$$

$$A_{0,i}\big[(1+\lambda_i)^{1-\gamma} - 1\big] = \Delta W_i, \qquad \Delta W_i = W_i^B - W_i^A, \tag{35}$$

$$(1+\lambda_i)^{1-\gamma} = 1 + \frac{\Delta W_i}{A_{0,i}}. \tag{36}$$

Raising to $1/(1-\gamma)$ and subtracting one recovers (20), $\lambda_i = (1+\Delta W_i/A_{0,i})^{1/(1-\gamma)} - 1$; since $x \mapsto x^{1-\gamma}$ is strictly monotone on $x > 0$, λ_i is unique.

Two checks First, when housing, ownership, and bequest carry no utility, $H_i = 0$ and $A_{0,i} = W_i^A$, and the formula collapses to $\lambda_i = (W_i^B/W_i^A)^{1/(1-\gamma)} - 1$, the total-utility ratio of Flöden (2001); our expression differs from it only in using the consumption sub-value $A_{0,i}$ in place of total value W_i^A. That replacement is not cosmetic here. With $\gamma > 1$ the power term is negative, so $A_{0,i} < 0$, whereas the additive housing, ownership, and normalisation terms make total value $W_i^A > 0$; the total-utility ratio $(W_i^B/W_i^A)^{1/(1-\gamma)}$ would then return the wrong sign, while the consumption sub-value $A_{0,i}$ delivers the correct one.